Probability: Concepts and Applications

Lecture notes, Lectures 1–10

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Lecture 1. Combinatorial Analysis

Many probability questions on finite sample spaces with equally likely outcomes reduce to counting. This lecture collects the counting tools used throughout the course.

1.1 The basic principle of counting

Motivation. For equally likely outcomes, a probability is a ratio of two counts. Most early problems are therefore counting problems, and they all reduce to one rule: multiply the number of choices at each stage.

Theorem 1.1 (Generalized counting principle)

If r experiments are performed in sequence so that the first has n_1 possible outcomes and, for each outcome of the first $i-1$ experiments, the i-th has n_i possible outcomes, then the r experiments together have $n_1 n_2 \cdots n_r$ possible outcomes.

Proof. For $r = 2$ list the outcomes as pairs (i, j), $1 \le i \le n_1$, $1 \le j \le n_2$: an $n_1 \times n_2$ array. Induct on r, treating the first $r-1$ experiments as one. ■

The number n_i may not depend on earlier outcomes, but the *set* of outcomes may.

Example 1.1 (License plates)

A plate has 3 letters followed by 4 digits. There are $26^3 \cdot 10^4 = 175{,}760{,}000$ plates. If no letter and no digit may repeat, there are $26 \cdot 25 \cdot 24 \cdot 10 \cdot 9 \cdot 8 \cdot 7 = 78{,}624{,}000$. The number of functions from an n-point set to $\{0, 1\}$ is 2^n, by the same principle.

1.2 Permutations

An ordered arrangement of n distinct objects is a **permutation**; there are $n! = n(n-1)\cdots 1$ of them, with $0! = 1$. The number of ordered selections of r objects out of n is $n(n-1)\cdots(n-r+1) = n!/(n-r)!$.

If the n objects fall into r groups of $n_1, \ldots, n_r$ mutually indistinguishable objects, each distinguishable arrangement corresponds to exactly $n_1! \cdots n_r!$ permutations of labelled objects (permute within each group), so there are

$$\frac{n!}{n_1!\, n_2! \cdots n_r!}$$

distinguishable arrangements.

Example 1.2

(a) The letters of PEPPER (three P, two E, one R) give $6!/(3!\,2!\,1!) = 60$ arrangements.
(b) Four mathematics, three chemistry, two history and one language book are shelved so that books on the same subject stand together. Order the four subject blocks (4! ways), then order within each block: $4!\,(4!\,3!\,2!\,1!) = 24 \cdot 288 = 6912$ arrangements.
(c) A class of 6 men and 4 women is ranked by an exam with no ties: $10! = 3{,}628{,}800$ rankings; if men are ranked only among men and women among women, $6!\,4! = 17{,}280$.

1.3 Combinations

The number of *unordered* groups of r objects chosen from n is

$$\binom{n}{r} = \frac{n(n-1)\cdots(n-r+1)}{r!} = \frac{n!}{(n-r)!\,r!}, \qquad 0 \le r \le n,$$

since each group is counted $r!$ times among the ordered selections. We set $\binom{n}{r} = 0$ for $r < 0$ or $r > n$. Clearly $\binom{n}{r} = \binom{n}{n-r}$.

Proposition 1.2 (Pascal's identity)

$$\binom{n}{r} = \binom{n-1}{r-1} + \binom{n-1}{r} \text{ for } 1 \le r \le n.$$

Proof. Fix one object. The r-subsets containing it number $\binom{n-1}{r-1}$; those not containing it number $\binom{n-1}{r}$. ∎

Theorem 1.3 (Binomial theorem)

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k}.$$

Proof. Expand $(x_1+y_1)\cdots(x_n+y_n)$: it is the sum of 2^n products, one for each choice of x_i or y_i from every factor. The products with exactly k of the x's are indexed by the k-subsets of $\{1,\dots,n\}$; there are $\binom{n}{k}$ of them. Now set $x_i = x$, $y_i = y$. ∎

Taking $x = y = 1$ shows that an n-set has $\sum_k \binom{n}{k} = 2^n$ subsets; $x = -1, y = 1$ gives $\sum_k (-1)^k \binom{n}{k} = 0$ for $n \ge 1$.

Combinatorial arguments ("count one set in two ways") prove many identities quickly.

Proposition 1.4

$$\text{(a)} \sum_{k=1}^{n} k\binom{n}{k} = n2^{n-1}, \qquad \text{(b)} \binom{n+m}{r} = \sum_{k=0}^{r} \binom{n}{k}\binom{m}{r-k} \quad \text{(Vandermonde)}.$$

Proof. (a) Count pairs (committee, chair) from n people. Choosing a committee of size k and then its chair gives the left side; choosing the chair first (n ways) and then any subset of the others (2^{n-1}) gives the right.

(b) Choose r people from n men and m women by first fixing the number k of men. ∎

Example 1.3 (Antennas in a row)

Of n antennas in a line, m are defective and indistinguishable, as are the $n-m$ working ones. How many orderings have no two defectives adjacent? Line up the $n-m$ working antennas; this creates $n-m+1$ gaps (including both ends), and at most one defective may go into each gap. So the answer is $\binom{n-m+1}{m}$. For instance, $n = 4, m = 2$ gives $\binom{3}{2} = 3$: DWDW, DWWD, WDWD.

1.4 Multinomial coefficients

The number of ways to divide n distinct objects into r *labelled* groups of sizes $n_1, \dots, n_r$ ($\sum n_i = n$) is

$$\binom{n}{n_1, \dots, n_r} = \binom{n}{n_1}\binom{n-n_1}{n_2}\cdots\binom{n-n_1-\cdots-n_{r-1}}{n_r} = \frac{n!}{n_1!\cdots n_r!},$$

which is the same number as in Section 1.2: assigning object j a group label is the same as arranging a word with n_i copies of label i.

Theorem 1.5 (Multinomial theorem)

$(x_1 + \cdots + x_r)^n = \sum_{n_1+\cdots+n_r=n} \binom{n}{n_1, \dots, n_r} x_1^{n_1}\cdots x_r^{n_r}$, the sum over nonnegative integer vectors.

Example 1.4 (Labelled versus unlabelled groups)

Ten children are split into an A team and a B team of five each: $\binom{10}{5,5} = 252$ ways. If the two teams are not labelled (just "two teams of five"), each division is counted twice, giving 126.

Example 1.5 (Knockout tournament)

In a knockout tournament with $n = 2^m$ players, the first round pairs the players and each pair produces one winner. The number of pairings is $n!/\{2^{n/2}(n/2)!\}$ (arrange in a row, pair neighbours, then forget the order within and among pairs), and each pairing has $2^{n/2}$ outcomes, so the first round has $n!/(n/2)!$ possible outcomes. Repeating round by round, the whole tournament has

$$\frac{n!}{(n/2)!}\cdot\frac{(n/2)!}{(n/4)!}\cdots\frac{2!}{1!} = n!$$

outcomes.

1.5 Integer solutions of equations

Motivation. Distributing identical objects into distinct boxes is a counting problem that appears in occupancy and urn models. The stars-and-bars argument turns it into choosing positions for separators.

Proposition 1.6 (Stars and bars)

There are $\binom{n-1}{r-1}$ vectors of *positive* integers with $x_1 + \cdots + x_r = n$, and $\binom{n+r-1}{r-1}$ vectors of *nonnegative* integers.

Proof. Write n stars in a row; a positive solution corresponds to choosing $r-1$ of the $n-1$ gaps between stars as dividers. For nonnegative solutions put $y_i = x_i + 1$: positive solutions of $\sum y_i = n + r$, of which there are $\binom{n+r-1}{r-1}$. ∎

Equivalently, $\binom{n+r-1}{r-1}$ counts the ways to put n indistinguishable balls into r distinguishable urns, and it is the number of terms in the multinomial expansion of $(x_1 + \cdots + x_r)^n$.

Example 1.6 (Investment)

\$20,000 is invested in units of \$1000 among 4 options. If all of it must be invested, there are $\binom{23}{3} = 1771$ strategies. If not all need be invested, add a fifth "cash" variable: $\binom{24}{4} = 10{,}626$. If each option must receive at least one unit, $\binom{19}{3} = 969$.

Example 1.7 (Antennas revisited)

In Example 1.3, an arrangement is determined by the numbers $x_1, \dots, x_{m+1}$ of working antennas before the first defective, between consecutive defectives, and after the last. These satisfy $\sum x_i = n - m$ with $x_1, x_{m+1} \geq 0$ and $x_2, \dots, x_m \geq 1$. Substituting $y_1 = x_1 + 1$, $y_{m+1} = x_{m+1} + 1$ gives a positive solution of $\sum y_i = n - m + 2$, so the count is $\binom{n-m+1}{m}$ again. Requiring at least two working antennas between defectives is now easy: substitute $y_i = x_i - 1$ for the interior terms.

1.6 Selected exercises

Motivation. These exercises are Ross's theoretical exercises for Chapter 1 (numbered as in the book). They train the main technique of the lecture, counting one set in two ways, and prove the multinomial theorem, which was stated above without proof.

Exercise 1.11 (Classifying subsets by their largest element)

Show that for $1 \leq k \leq n$,

$$\sum_{i=k}^{n} \binom{i-1}{k-1} = \binom{n}{k}.$$

Solution.

Step 1. Let $\mathcal{A}$ be the family of k-element subsets of $\{1, \dots, n\}$, so $|\mathcal{A}| = \binom{n}{k}$. Every $A \in \mathcal{A}$ has a unique largest element $\max A$, and $\max A \geq k$ because A contains k distinct positive integers. Hence $\mathcal{A}$ is the disjoint union of the families $\mathcal{A}_i = \{A \in \mathcal{A} : \max A = i\}$, $i = k, \dots, n$.

Step 2. A set $A \in \mathcal{A}_i$ is determined by its other $k-1$ elements, which form an arbitrary $(k-1)$-subset of $\{1, \dots, i-1\}$; conversely every such subset together with i lies in $\mathcal{A}_i$. So $|\mathcal{A}_i| = \binom{i-1}{k-1}$.

Step 3. Adding over i gives $\binom{n}{k} = \sum_{i=k}^{n} |\mathcal{A}_i| = \sum_{i=k}^{n} \binom{i-1}{k-1}$. ■

Exercise 1.12 (Committees with a chair and a secretary)

Show that for $n \geq 1$,

$$\sum_{k=1}^{n} k^2 \binom{n}{k} = 2^{n-2} n(n+1).$$

(The identity $\sum_k k\binom{n}{k} = n2^{n-1}$ is Proposition 1.4(a).)

Solution. From n people we count the triples (C, a, b) where C is a nonempty committee, $a \in C$ is its chair and $b \in C$ its secretary, $a = b$ being allowed.

Step 1. (Committee first.) There are $\binom{n}{k}$ committees of size k, and for each of them k choices of a and k of b. So the number of triples is $\sum_{k=1}^{n} k^2 \binom{n}{k}$.

Step 2. (Officers first, $a = b$.) Choose the person $a = b$ in n ways; each of the other $n - 1$ people is in or out of C freely. This gives $n2^{n-1}$ triples.

Step 3. (Officers first, $a \neq b$.) Choose the ordered pair (a, b) in $n(n-1)$ ways and the rest of C among the other $n-2$ people in 2^{n-2} ways, giving $n(n-1)2^{n-2}$ triples (this is 0 when $n = 1$).

Step 4. Steps 2 and 3 count disjoint classes whose union is the set counted in Step 1, so

$$\sum_{k=1}^{n} k^2 \binom{n}{k} = n2^{n-1} + n(n-1)2^{n-2} = 2^{n-2}(2n + n^2 - n) = 2^{n-2}n(n+1).$$

■

Exercise 1.16 (Tournaments with ties)

In a tournament of n contestants the outcome is an ordering of the players with ties allowed: the players are divided into groups, the first group being those tied for first place, the next group those tied for the next place, and so on. Let $N(n)$ be the number of outcomes, with $N(0) = 1$.
(a) Show that $N(n) = \sum_{i=1}^{n} \binom{n}{i} N(n-i)$ for $n \geq 1$.
(b) Show that this is equivalent to $N(n) = \sum_{i=0}^{n-1} \binom{n}{i} N(i)$.
(c) Find $N(3)$ and $N(4)$.

Solution.

(a) Classify the outcomes by the number $i \in \{1, \ldots, n\}$ of players tied for last place. An outcome with i players last is determined by the set L of those players and by the outcome among the remaining $n-i$ players (which is an arbitrary ordering with ties of them; if $i = n$ there is exactly one, matching $N(0) = 1$). Conversely any set L of size i and any outcome of the others, with L appended as the last group, is an outcome with exactly i players last. This bijection gives $\binom{n}{i} N(n-i)$ outcomes with i players last, and summing over i proves the claim.

(b) Substitute $j = n - i$, which runs over $0, \ldots, n-1$, and use $\binom{n}{n-j} = \binom{n}{j}$: $\sum_{i=1}^{n} \binom{n}{i} N(n-i) = \sum_{j=0}^{n-1} \binom{n}{j} N(j)$.

(c) From (b), $N(1) = N(0) = 1$, $N(2) = N(0) + 2N(1) = 3$,

$$N(3) = N(0) + 3N(1) + 3N(2) = 1 + 3 + 9 = 13,$$
$$N(4) = N(0) + 4N(1) + 6N(2) + 4N(3) = 1 + 4 + 18 + 52 = 75.$$

■

Exercise 1.19 (The multinomial theorem)

Prove Theorem 1.5.

Solution.

Step 1. Write $(x_1 + \cdots + x_r)^n$ as a product of n factors and expand by the distributive law. The result is the sum, over all words $w = (i_1, \ldots, i_n) \in \{1, \ldots, r\}^n$, of the products $x_{i_1} x_{i_2} \cdots x_{i_n}$: the word records which summand is taken from each factor, and different words are different terms of the expansion.

Step 2. If n_j is the number of positions t with $i_t = j$, then by commutativity $x_{i_1} \cdots x_{i_n} = x_1^{n_1} \cdots x_r^{n_r}$, and $(n_1, \ldots, n_r)$ is a vector of nonnegative integers with sum n. Grouping the words by this vector,

$$(x_1 + \cdots + x_r)^n = \sum_{n_1 + \cdots + n_r = n} c(n_1, \ldots, n_r)\, x_1^{n_1} \cdots x_r^{n_r},$$

where $c(n_1, \ldots, n_r)$ is the number of words of length n containing exactly n_j copies of the letter j for each j.

Step 3. Such a word is an arrangement of n letters, n_j of which are alike of kind j; by Section 1.2 there are $n!/(n_1! \cdots n_r!) = \binom{n}{n_1,\ldots,n_r}$ of them. Hence $c(n_1, \ldots, n_r) = \binom{n}{n_1,\ldots,n_r}$, which is the theorem. ■

Lecture 2. Axioms of Probability

2.1 Sample space and events

The **sample space** S is the set of all possible outcomes of an experiment; an **event** is a subset of S. For events E, F we use the union $E \cup F$, the intersection $EF = E \cap F$, the complement $E^c = S \setminus E$, and the empty event $\emptyset$. E and F are **mutually exclusive** if $EF = \emptyset$. Union and intersection are commutative, associative and distribute over each other, and

$$\Big(\bigcup_{i=1}^{n} E_i\Big)^c = \bigcap_{i=1}^{n} E_i^c, \qquad \Big(\bigcap_{i=1}^{n} E_i\Big)^c = \bigcup_{i=1}^{n} E_i^c \qquad \text{(De Morgan)}.$$

For the first, an outcome lies outside every E_i exactly when it lies outside their union; the second follows by applying the first to the E_i^c.

2.2 Axioms of probability

The relative-frequency idea, $P(E) = \lim n(E)/n$ over repeated trials, motivates but does not define probability; instead we impose axioms.

Definition (Probability)

A function P on events is a **probability** if (i) $0 \le P(E) \le 1$;
(ii) $P(S) = 1$;
(iii) for mutually exclusive $E_1, E_2, \ldots$, $P\Big(\bigcup_{i=1}^{\infty} E_i\Big) = \sum_{i=1}^{\infty} P(E_i)$.

Taking $E_1 = S$ and $E_i = \emptyset$ for $i \ge 2$ forces $P(\emptyset) = 0$, and then (iii) gives finite additivity. (For uncountable S one can only define P on a suitable class of events; we ignore this.)

Proposition 2.1

(a) $P(E^c) = 1 - P(E)$.
(b) $E \subset F \Rightarrow P(E) \le P(F)$.
(c) $P(E \cup F) = P(E) + P(F) - P(EF)$.

Proof. (a) E and E^c are disjoint with union S.

(b) $F = E \cup E^c F$, a disjoint union.

(c) $E \cup F = E \cup E^c F$ and $F = EF \cup E^c F$, both disjoint; subtract. ∎

Theorem 2.2 (Inclusion–exclusion)

$$P\Big(\bigcup_{i=1}^{n} E_i\Big) = \sum_i P(E_i) - \sum_{i<j} P(E_i E_j) + \ldots + (-1)^{r+1} \sum_{i_1<\cdots<i_r} P(E_{i_1} \cdots E_{i_r}) + \ldots + (-1)^{n+1} P(E_1 \cdots E_n).$$

Proof. For a finite or countable S it suffices to check that each outcome contributes its probability equally to both sides. An outcome in none of the E_i contributes 0 to both. An outcome in exactly $m \geq 1$ of them contributes once on the left, and on the right $\binom{m}{1} - \binom{m}{2} + \cdots \pm \binom{m}{m} = 1 - \sum_{i=0}^{m} (-1)^i \binom{m}{i} = 1$ times. (For general S, induct on n using Proposition 2.1(c).) ■

Truncating the sum gives alternately upper and lower bounds (Bonferroni): $P(\bigcup E_i) \leq \sum P(E_i)$ and $P(\bigcup E_i) \geq \sum P(E_i) - \sum_{i<j} P(E_i E_j)$, and so on.

2.3 Equally likely outcomes

If $S = \{1, \ldots, N\}$ and every outcome has probability $1/N$, then $P(E) = |E|/N$, and probability reduces to Lecture 1. It is often convenient to treat objects as distinguishable even when they are not; this never changes the answer and makes outcomes equally likely.

Example 2.1

(a) Three balls are drawn from an urn with 6 white and 5 black balls. Then $P(\text{one white, two black}) = \binom{6}{1}\binom{5}{2}/\binom{11}{3} = 60/165 = 4/11$. Counting ordered draws gives $3 \cdot 6 \cdot 5 \cdot 4/(11 \cdot 10 \cdot 9) = 4/11$ as well.
(b) k of n balls are drawn, one of them special. By symmetry, the special ball is equally likely to be in any position of a random ordering, so $P(\text{special drawn}) = k/n$; directly, $\binom{1}{1}\binom{n-1}{k-1}/\binom{n}{k} = k/n$.
(c) A poker full house: choose the rank of the triple and its 3 suits, then the rank of the pair and its 2 suits: $13\binom{4}{3} \cdot 12\binom{4}{2}/\binom{52}{5} = 3744/2{,}598{,}960 \approx 0.00144$.

Example 2.2 (Birthday problem)

Ignoring leap years and assuming all 365^n birthday assignments equally likely, the probability that n people have distinct birthdays is $365 \cdot 364 \cdots (366 - n)/365^n$. This drops below $1/2$ at $n = 23$: the probability of a shared birthday is 0.507 for $n = 23$ and 0.970 for $n = 50$. The surprise disappears when one counts pairs: 23 people form $\binom{23}{2} = 253$ pairs, each matching with probability $1/365$.

Example 2.3 (The card after the first ace)

A shuffled deck is turned up card by card. Is the card following the first ace more likely to be the ace of spades or the two of clubs? Count orderings in which the ace of spades follows the first ace: remove it, order the other 51 cards (51! ways), and reinsert it right after the first ace. So the probability is $51!/52! = 1/52$. The same argument applies to the two of clubs, so both have probability $1/52$.

Example 2.4 (The matching problem)

N people throw their hats into a pile and each picks one at random. Let E_i be the event that person i gets their own hat. For $i_1 < \cdots < i_r$, $P(E_{i_1} \cdots E_{i_r}) = (N - r)!/N!$, and there are $\binom{N}{r}$ such terms, so by inclusion–exclusion

$$P\Big(\bigcup_i E_i\Big) = \sum_{r=1}^{N} (-1)^{r+1} \binom{N}{r} \frac{(N-r)!}{N!} = \sum_{r=1}^{N} \frac{(-1)^{r+1}}{r!},$$

so that

$$P(\text{no match}) = \sum_{r=0}^{N} \frac{(-1)^r}{r!} \xrightarrow[N\to\infty]{} e^{-1} \approx 0.368.$$

Exactly k matches: choose the k people who match and require no match among the other $N-k$; this gives

$$P(\text{exactly } k) = \binom{N}{k} \frac{(N-k)!}{N!} \sum_{r=0}^{N-k} \frac{(-1)^r}{r!} = \frac{1}{k!} \sum_{r=0}^{N-k} \frac{(-1)^r}{r!} \approx \frac{e^{-1}}{k!},$$

a Poisson(1) probability (Lecture 4).

Example 2.5 (Couples at a round table)

Ten couples sit at random around a round table (19! equally likely circular orders). Let E_i be the event that couple i sit together. Gluing r given couples into blocks leaves $20-r$ units, which can be arranged in $(19-r)!$ circular orders, and each block in 2 ways. Hence

$$P(\text{no couple together}) = \sum_{r=0}^{10} (-1)^r \binom{10}{r} \frac{2^r (19-r)!}{19!} \approx 0.3395.$$

2.4 Probability as a continuous set function

Motivation. Many events are limits of simpler ones, for example "eventually" or "infinitely often". Countable additivity implies that probability respects such monotone limits, which is what lets us compute probabilities of infinite events.

A sequence of events is **increasing** if $E_n \subset E_{n+1}$, with limit $\bigcup E_n$, and **decreasing** if $E_n \supset E_{n+1}$, with limit $\bigcap E_n$.

Proposition 2.3 (Continuity)

For an increasing or decreasing sequence, $\lim_n P(E_n) = P(\lim_n E_n)$.

Proof. Increasing case: put $F_1 = E_1$ and $F_n = E_n E_{n-1}^c$. The F_n are disjoint, $\bigcup_1^n F_i = E_n$ and $\bigcup_1^\infty F_i = \bigcup_1^\infty E_i$, so by axiom (iii)

$$P\Big(\bigcup_i E_i\Big) = \sum_{i=1}^{\infty} P(F_i) = \lim_n \sum_{i=1}^{n} P(F_i) = \lim_n P(E_n).$$

Decreasing case: apply this to the increasing E_n^c and use De Morgan. ■

Example 2.6

A fair coin is tossed forever. Let E_n be the event that the first n tosses are all tails; then E_n decreases to the event T that no head ever appears, so $P(T) = \lim 2^{-n} = 0$. Thus a head appears eventually with probability 1, although "all tails" is a possible outcome: probability zero does not mean impossible.

Subjective probability (a person's degree of belief) obeys the same axioms if beliefs are to be coherent, so the theory applies to it as well.

2.5 Selected exercises

Motivation. These are Ross's theoretical exercises for Chapter 2 (book numbering). They supply the induction proof of inclusion–exclusion for a general sample space, a Bonferroni bound, and two standard counting and modelling facts.

Exercise 2.8 (Counting partitions)

Let T_n be the number of partitions of $\{1,\dots,n\}$ into nonempty blocks, and put $T_0 = 1$.
(a) Show that $T_3 = 5$.
(b) Show that $T_{n+1} = 1 + \sum_{k=1}^{n} \binom{n}{k} T_k$ for $n \geq 1$, and deduce $T_4 = 15$.

Solution.

(a) The partitions of $\{1,2,3\}$ are $\{123\}$, $\{1\}\{23\}$, $\{2\}\{13\}$, $\{3\}\{12\}$ and $\{1\}\{2\}\{3\}$.

(b)

Step 1. Call the element $n+1$ special. A partition π of $\{1,\dots,n+1\}$ determines the block $B \ni n+1$ and the partition π' of the remaining set $R = \{1,\dots,n\} \setminus B$ formed by the other blocks (π' is empty if $R = \emptyset$). Conversely, a subset $A \subseteq \{1,\dots,n\}$ and a partition π' of $\{1,\dots,n\} \setminus A$ give back the partition $\{A \cup \{n+1\}\} \cup \pi'$. These maps are inverse to each other.

Step 2. Classify by $k = |R| \in \{0,\dots,n\}$. There are $\binom{n}{n-k} = \binom{n}{k}$ choices of A with $|A| = n-k$, and then T_k choices of π' when $k \geq 1$, and exactly one (the empty partition) when $k = 0$. Hence

$$T_{n+1} = 1 + \sum_{k=1}^{n} \binom{n}{k} T_k.$$

Step 3. With $T_1 = 1$, $T_2 = 2$ and $T_3 = 5$, $T_4 = 1 + 3T_1 + 3T_2 + T_3 = 1+3+6+5 = 15$. Continuing, $T_5 = 52$, $T_6 = 203$, …, $T_{10} = 115{,}975$. ∎

Exercise 2.14 (Inclusion–exclusion by induction)

Prove Theorem 2.2 for an arbitrary sample space by induction on n.

Solution. For $\emptyset \neq I \subseteq \{1,\dots,n\}$ write $E_I = \bigcap_{i \in I} E_i$. The claim is $P(\bigcup_{i=1}^{n} E_i) = \sum_I (-1)^{|I|+1} P(E_I)$, the sum over nonempty $I \subseteq \{1,\dots,n\}$.

Step 1. For $n = 1$ both sides equal $P(E_1)$; for $n = 2$ the claim is Proposition 2.1(c).

Step 2. Assume the claim for every family of n events and let $E_1,\dots,E_{n+1}$ be given. Put $U = \bigcup_{i=1}^{n} E_i$. By Proposition 2.1(c) and $UE_{n+1} = \bigcup_{i=1}^{n} (E_i E_{n+1})$,

$$P\Big(\bigcup_{i=1}^{n+1} E_i\Big) = P(U) + P(E_{n+1}) - P\Big(\bigcup_{i=1}^{n} E_i E_{n+1}\Big).$$

Step 3. Apply the induction hypothesis to $E_1,\dots,E_n$ and to $E_1E_{n+1},\dots,E_nE_{n+1}$. For $\emptyset \neq J \subseteq \{1,\dots,n\}$ the intersection of the events E_jE_{n+1}, $j \in J$, is $E_{J \cup \{n+1\}}$, so

$$P\Big(\bigcup_{i=1}^{n+1} E_i\Big) = \sum_J (-1)^{|J|+1} P(E_J) + P(E_{\{n+1\}}) - \sum_J (-1)^{|J|+1} P(E_{J \cup \{n+1\}}),$$

both sums over nonempty $J \subseteq \{1,\dots,n\}$.

Step 4. Every nonempty $I \subseteq \{1, \dots, n+1\}$ occurs exactly once on the right: if $n+1 \notin I$, in the first sum with sign $(-1)^{|I|+1}$; if $I = \{n+1\}$, as the middle term with sign $+1 = (-1)^{1+1}$; if $I = J \cup \{n+1\}$ with $J \neq \emptyset$, in the last sum with sign $-(-1)^{|J|+1} = (-1)^{|I|+1}$ since $|I| = |J| + 1$. This is the claim for $n+1$ events. ■

Exercise 2.16 (Bonferroni's inequality)

Show that for any events $E_1, \dots, E_n$,

$$P(E_1 E_2 \cdots E_n) \geq P(E_1) + \cdots + P(E_n) - (n-1).$$

Solution.

Step 1. ($n = 2$.) By Proposition 2.1(c) and $P(E \cup F) \leq 1$, $P(EF) = P(E) + P(F) - P(E \cup F) \geq P(E) + P(F) - 1$. For $n = 1$ the claim is an equality.

Step 2. (Induction.) If the claim holds for n events, apply Step 1 to $E = E_1 \cdots E_n$ and $F = E_{n+1}$:

$$\begin{aligned} P(E_1 \cdots E_{n+1}) &\geq P(E_1 \cdots E_n) + P(E_{n+1}) - 1 \\ &\geq \sum_{i=1}^{n} P(E_i) - (n-1) + P(E_{n+1}) - 1 = \sum_{i=1}^{n+1} P(E_i) - n. \end{aligned}$$

For example, if $P(E) = 0.9$ and $P(F) = 0.8$, then $P(EF) \geq 0.7$ whatever the dependence between E and F. ■

Exercise 2.19 (Waiting for the r-th red ball)

An urn contains n red and m blue balls. They are withdrawn one at a time, at random, until a total of $r \leq n$ red balls have been withdrawn. Find the probability that exactly k balls are withdrawn.

Solution.

Step 1. Label the balls and take as sample space the $(n+m)!$ equally likely orders in which all of them would be withdrawn. Exactly k balls are withdrawn iff the first $k-1$ positions contain exactly $r-1$ red balls and position k holds a red ball; this requires $r \leq k \leq r+m$.

Step 2. Count such orders: choose the red and blue balls in the first $k-1$ positions ($\binom{n}{r-1}\binom{m}{k-r}$ ways) and their order ($(k-1)!$ ways), the red ball in position k among the $n-r+1$ remaining red balls, and the order of the last $n+m-k$ balls ($(n+m-k)!$ ways).

Step 3. Dividing by $(n+m)!$ and using $\binom{n+m}{k-1}^{-1} = (k-1)!\,(n+m-k+1)!/(n+m)!$,

$$P(k \text{ balls withdrawn}) = \frac{\binom{n}{r-1}\binom{m}{k-r}}{\binom{n+m}{k-1}} \cdot \frac{n-r+1}{n+m-k+1}, \qquad r \leq k \leq r+m.$$

The first factor is the hypergeometric probability of $r-1$ red balls among the first $k-1$ draws, and the second the probability that the next draw is red given this. ■

Exercise 2.20 (No uniform distribution on a countable set)

Let the sample space $S = \{s_1, s_2, \dots\}$ be countably infinite. Show that not all outcomes can be equally likely. Can all outcomes have positive probability?

Solution.

Step 1. The singletons $\{s_i\}$ are mutually exclusive with union S, so by axioms (ii) and (iii), $1 = P(S) = \sum_{i=1}^{\infty} P(\{s_i\})$.

Step 2. If $P(\{s_i\}) = p$ for every i, the series is 0 when $p = 0$ and $+\infty$ when $p > 0$; neither equals 1. So the outcomes cannot all be equally likely.

Step 3. All outcomes can have positive probability: $P(\{s_i\}) = 2^{-i}$, extended to every event E by $P(E) = \sum_{i:s_i \in E} 2^{-i}$, is nonnegative, has $P(S) = \sum_i 2^{-i} = 1$, and is countably additive because a series of nonnegative terms may be summed in any grouping. ■

Lecture 3. Conditional Probability and Independence

3.1 Conditional probability

Motivation. New information changes probabilities. Conditional probability restricts the sample space to what we know has occurred, and the multiplication rule lets us build complex probabilities step by step.

Definition (Conditional probability)

If $P(F) > 0$, the conditional probability of E given F is $P(E \mid F) = P(EF)/P(F)$.

With equally likely outcomes, conditioning on F makes the outcomes in F equally likely, so $P(E \mid F) = |EF|/|F|$: we simply shrink the sample space to F.

Proposition 3.1 (Multiplication rule)

$$P(E_1E_2\cdots E_n) = P(E_1)P(E_2 \mid E_1)P(E_3 \mid E_1E_2)\cdots P(E_n \mid E_1\cdots E_{n-1}).$$

Proof. Write each factor as a ratio; the product telescopes. ∎

Example 3.1

(a) An urn holds 8 red and 4 white balls; two are drawn without replacement. $P(\text{both red}) = \frac{8}{12}\cdot\frac{7}{11} = \frac{14}{33}$, which agrees with $\binom{8}{2}/\binom{12}{2}$.
(b) A deck is divided at random into four piles of 13. Let E_i be the event that the i-th pile has exactly one ace (and put the aces in piles in order). Placing the aces one at a time among the remaining slots,

$$P(E_1E_2E_3E_4) = 1\cdot\frac{39}{51}\cdot\frac{26}{50}\cdot\frac{13}{49} \approx 0.105,$$

since given that the first ace is in some pile, the second ace must be in one of the 39 of the remaining 51 positions that lie in other piles, and so on.

Example 3.2 (Matching, recursively)

In the matching problem (Example 2.4), let P_N be the probability of no match among N people and M the event that person 1 takes their own hat. Then $P_N = P(\text{no match} \mid M^c)\frac{N-1}{N}$. Given M^c, say person 1 took hat $j \neq 1$. The other $N-1$ people choose among $N-1$ hats that include hat 1 but not hat j. Either person j takes hat 1 (probability $\frac{1}{N-1}$) and the remaining $N-2$ must have no match, or we may regard hat 1 as person j's "own" hat, and then we need no match among $N-1$ people. So

$$P_N = \frac{N-1}{N}\Big(P_{N-1} + \frac{P_{N-2}}{N-1}\Big), \qquad \text{i.e.} \qquad P_N - P_{N-1} = -\frac{1}{N}(P_{N-1} - P_{N-2}).$$

With $P_1 = 0$, $P_2 = \frac{1}{2}$, induction gives $P_N - P_{N-1} = (-1)^N/N!$ and hence $P_N = \sum_{r=0}^{N}(-1)^r/r!$ once more.

3.2 Total probability and Bayes's formula

Motivation. Often we know $P(B \mid A_i)$ but want $P(A_i \mid B)$, for instance the probability of a disease given a positive test. Conditioning on the cases A_i and inverting gives Bayes's formula, whose answers are frequently counterintuitive.

Since $E = EF \cup EF^c$,

$$P(E) = P(E \mid F)P(F) + P(E \mid F^c)(1 - P(F)),$$

a weighted average of the conditional probabilities. More generally:

Theorem 3.2 (Total probability and Bayes's formula)

If $F_1, \dots, F_n$ are mutually exclusive with union S, then

$$P(E) = \sum_{i=1}^{n} P(E \mid F_i)P(F_i), \qquad P(F_j \mid E) = \frac{P(E \mid F_j)P(F_j)}{\sum_i P(E \mid F_i)P(F_i)}.$$

The $P(F_j)$ are **prior** probabilities of the hypotheses, and $P(F_j \mid E)$ are **posterior** probabilities after the evidence E. In odds form, for a hypothesis H,

$$\frac{P(H \mid E)}{P(H^c \mid E)} = \frac{P(H)}{P(H^c)} \cdot \frac{P(E \mid H)}{P(E \mid H^c)} :$$

new evidence multiplies the odds by the **likelihood ratio**.

Example 3.3 (A screening test)

A test detects a disease with probability 0.95 when it is present, but gives a false positive for 1% of healthy people. If 0.5% of the population has the disease,

$$P(D \mid +) = \frac{0.95 \cdot 0.005}{0.95 \cdot 0.005 + 0.01 \cdot 0.995} \approx 0.323.$$

Only a third of positives are diseased: the healthy are so numerous that their 1% of false positives outnumbers the true positives.

Example 3.4 (Multiple choice)

A student knows the answer with probability p and otherwise guesses among m alternatives. Then

$$P(\text{knew} \mid \text{correct}) = \frac{p}{p + (1-p)/m} = \frac{mp}{1 + (m-1)p}.$$

With $m = 5$, $p = 1/2$ this is $5/6$.

Example 3.5 (Three cards)

One card is red on both sides, one black on both sides, one red on one side and black on the other. A card is drawn at random and laid down; the upper side is red. The tempting answer $1/2$ for "the other side is red" is wrong: there are three equally likely red faces that could be on top, and two of them belong to the red–red card. Formally $P(RR \mid R \text{ up}) = \frac{1/3}{1/3 \cdot 1 + 1/3 \cdot 1/2} = \frac{2}{3}$.

Example 3.6 (Searching for a plane)

A plane is equally likely to be in one of three regions. If it is in region i, a search there misses it with probability β_i. After an unsuccessful search of region 1,

$$P(R_1 \mid \text{miss}) = \frac{\beta_1/3}{\beta_1/3 + 1/3 + 1/3} = \frac{\beta_1}{\beta_1 + 2}, \qquad P(R_j \mid \text{miss}) = \frac{1}{\beta_1 + 2} \quad (j = 2, 3).$$

The probability of region 1 decreases and the others increase, as intuition says.

For a fixed F with $P(F) > 0$, $Q(E) = P(E \mid F)$ is itself a probability, so all earlier results apply to it; e.g. $P(E_1 \mid F) = P(E_1 \mid E_2F)P(E_2 \mid F) + P(E_1 \mid E_2^cF)P(E_2^c \mid F)$.

3.3 Independent events

Definition (Independence)

E and F are **independent** if $P(EF) = P(E)P(F)$. Events $E_1, \ldots, E_n$ are (mutually) independent if $P(E_{i_1} \cdots E_{i_r}) = P(E_{i_1}) \cdots P(E_{i_r})$ for every subcollection.

If E and F are independent then so are E and F^c, since $P(EF^c) = P(E) - P(E)P(F) = P(E)P(F^c)$.

Example 3.7 (Pairwise does not imply mutual)

Roll two fair dice. Let E: the sum is 7; F: the first die is 4; G: the second die is 3. Each pair is independent ($P(EF) = 1/36 = P(E)P(F)$ etc.), but $P(E \mid FG) = 1 \neq P(E)$. Note also that F_6 (sum 6) is *not* independent of F while E is: $P(\text{sum} = 7 \mid \text{first} = k) = 1/6$ for every k.

Example 3.8 (Reliability)

A parallel system works if at least one of n independent components works; component i works with probability p_i. Then $P(\text{works}) = 1 - \prod_i (1 - p_i)$. A series system works with probability $\prod_i p_i$.

Example 3.9 (E before F)

Independent trials are repeated; on each, E occurs with probability $P(E)$ and F with $P(F)$, $EF = \emptyset$. Conditioning on the first trial, $P(E \text{ first}) = P(E) + (1 - P(E) - P(F))P(E \text{ first})$, so

$$P(E \text{ occurs before } F) = \frac{P(E)}{P(E) + P(F)}.$$

In craps, a first roll of 7 or 11 wins, 2, 3 or 12 loses, and any other sum k becomes the "point", after which one rolls until k (win) or 7 (lose). With $p_k = P(\text{sum} = k)$,

$$P(\text{win}) = p_7 + p_{11} + \sum_{k \in \{4,5,6,8,9,10\}} p_k \frac{p_k}{p_k + p_7} = \frac{244}{495} \approx 0.493.$$

Example 3.10 (The problem of the points)

Trials succeed independently with probability p. What is the probability of n successes before m failures? This happens iff there are at least n successes among the first $m + n - 1$ trials

(either way the contest is decided by then), so

$$P_{n,m} = \sum_{k=n}^{m+n-1} \binom{m+n-1}{k} p^k (1-p)^{m+n-1-k}.$$

This was Pascal and Fermat's question of dividing the stakes of an interrupted game: the stakes should be split in the ratio $P_{n,m} : 1 - P_{n,m}$. For instance with $p = 1/2$, $n = 4$, $m = 7$ it is 0.828.

Example 3.11 (Gambler's ruin)

A gambler with fortune i wins or loses one unit on each independent play, with probabilities p and $q = 1-p$, until reaching N or 0. Let P_i be the probability of reaching N. Conditioning on the first play, $P_i = pP_{i+1} + qP_{i-1}$, i.e. $P_{i+1} - P_i = \frac{q}{p}(P_i - P_{i-1})$, with $P_0 = 0$. Hence $P_{i+1} - P_i = (q/p)^i P_1$, and summing,

$$P_i = \begin{cases} \dfrac{1-(q/p)^i}{1-(q/p)^N}, & p \neq \frac{1}{2}, \\ \dfrac{i}{N}, & p = \frac{1}{2}, \end{cases}$$

after using $P_N = 1$ to find P_1. Letting $N \to \infty$: if $p \leq \frac{1}{2}$ the gambler is ruined with probability 1 against an infinitely rich opponent; if $p > \frac{1}{2}$ survival has probability $1 - (q/p)^i$.

3.4 $P(\cdot \mid F)$ is a probability

Since conditional probabilities obey the axioms, one may condition sequentially, and conditional independence makes sense: E_1, E_2 are conditionally independent given F if $P(E_1E_2 \mid F) = P(E_1 \mid F)P(E_2 \mid F)$.

Example 3.12 (Laplace's rule of succession)

A box contains $k+1$ coins; coin i shows heads with probability i/k, $i = 0, \ldots, k$. A coin is chosen at random and flipped n times, giving n heads. Given the coin, flips are independent, so

$$P(\text{head on flip } n+1 \mid n \text{ heads}) = \frac{\sum_i (i/k)^{n+1}/(k+1)}{\sum_i (i/k)^n/(k+1)} \approx \frac{\int_0^1 x^{n+1}dx}{\int_0^1 x^n dx} = \frac{n+1}{n+2}$$

for large k. The flips are not independent unconditionally: early heads are evidence that the coin is good.

Example 3.13 (Updating sequentially)

Hypotheses $H_1, \ldots, H_n$ have priors $P(H_i)$, and pieces of evidence $E_1, \ldots, E_k$ are conditionally independent given each H_i. Then

$$P(H_i \mid E_1 \cdots E_k) \propto P(H_i) \prod_{j=1}^{k} P(E_j \mid H_i),$$

so the posterior after E_1 serves as the prior for E_2, and so on; the order in which the evidence arrives does not matter.

3.5 Selected exercises

Motivation. These are Ross's theoretical exercises for Chapter 3 (book numbering). Each one is solved by the main idea of the lecture: condition on a well-chosen first or last event and obtain a recursion or a symmetry.

Exercise 3.7(a) (The colour that survives)

An urn contains $n \geq 1$ white and $m \geq 1$ black balls. The balls are withdrawn one at a time, at random, until only balls of one colour are left. Show that they are all white with probability $n/(n+m)$.

Solution.

Step 1. Imagine that the withdrawals continue until the urn is empty, and label the balls. All $(n+m)!$ orders of withdrawal are equally likely, and the actual experiment is the initial part of this order up to the stopping time.

Step 2. When we stop, the balls left in the urn are those in a final segment of the order. This segment is nonempty, because at the start both colours are present and so we never stop before at least one ball remains. Hence it contains the last ball of the order, and all the balls left have the colour of that last ball. So the balls left are white iff the last ball of the full order is white.

Step 3. For each fixed ball, exactly $(n+m-1)!$ orders end with it, so each ball is last with probability $1/(n+m)$. With n white balls, $P(\text{last ball white}) = n/(n+m)$. ∎

Exercise 3.16 (An even number of successes)

Let P_n be the probability that n Bernoulli trials with success probability p result in an even number of successes (0 counts as even). Show that

$$P_n = p(1 - P_{n-1}) + (1-p)P_{n-1} \quad (n \geq 1),$$

and deduce $P_n = (1 + (1-2p)^n)/2$.

Solution.

Step 1. Let S be the number of successes in the first $n-1$ trials. The total is even iff either trial n succeeds and S is odd, or trial n fails and S is even. These two events are disjoint, and trial n is independent of the first $n-1$ trials, so

$$P_n = p\,P(S \text{ odd}) + (1-p)P(S \text{ even}) = p(1 - P_{n-1}) + (1-p)P_{n-1}.$$

Step 2. The recursion reads $P_n = p + (1-2p)P_{n-1}$. For $n = 0$ there are no successes, so $P_0 = 1 = (1 + (1-2p)^0)/2$. If $P_{n-1} = (1 + (1-2p)^{n-1})/2$, then

$$P_n = p + \frac{1-2p}{2} + \frac{(1-2p)^n}{2} = \frac{1 + (1-2p)^n}{2},$$

which completes the induction. ∎

Exercise 3.18 (No run of three heads)

Let Q_n be the probability that no run of three consecutive heads appears in n tosses of a fair coin. Show that

$$Q_n = \tfrac{1}{2}Q_{n-1} + \tfrac{1}{4}Q_{n-2} + \tfrac{1}{8}Q_{n-3} \quad (n \geq 3), \qquad Q_0 = Q_1 = Q_2 = 1,$$

and find Q_8.

Solution.

Step 1. Let R be the event of no run of three heads and F_j $(j = 1, 2, 3)$ the event that the first tail occurs at toss j; let H be the event that tosses $1, 2, 3$ are all heads. Then F_1, F_2, F_3, H partition the sample space and $P(F_j) = 2^{-j}$.

Step 2. On H there is a run, so $RH = \emptyset$. For $j \leq 3$, on F_j the first j tosses are $j-1$ heads followed by a tail and contain no run; a run of three heads cannot straddle toss j, which is a tail. Hence $RF_j = F_j \cap R_j$, where R_j is the event of no run of three heads among tosses $j+1, \ldots, n$.

Step 3. R_j depends only on tosses $j+1, \ldots, n$, so it is independent of F_j, and these $n-j$ tosses are again independent fair tosses, so $P(R_j) = Q_{n-j}$. Therefore

$$Q_n = \sum_{j=1}^{3} P(F_j)P(R_j) = \tfrac{1}{2}Q_{n-1} + \tfrac{1}{4}Q_{n-2} + \tfrac{1}{8}Q_{n-3}.$$

Step 4. Iterating, $Q_3 = \frac{7}{8}$, $Q_4 = \frac{13}{16}$, $Q_5 = \frac{24}{32}$, $Q_6 = \frac{44}{64}$, $Q_7 = \frac{81}{128}$ and

$$Q_8 = \tfrac{1}{2}\cdot\tfrac{81}{128} + \tfrac{1}{4}\cdot\tfrac{44}{64} + \tfrac{1}{8}\cdot\tfrac{24}{32} = \frac{81+44+24}{256} = \frac{149}{256} \approx 0.582.$$

■

Exercise 3.21 (The ballot problem)

In an election, candidate A receives n votes and candidate B receives m votes, $n > m \geq 0$, and all $\binom{n+m}{n}$ orderings of the votes are equally likely. Let $P_{n,m}$ be the probability that A is strictly ahead throughout the count.
(a) Derive a recursion for $P_{n,m}$ in terms of $P_{n-1,m}$ and $P_{n,m-1}$ by conditioning on who receives the last vote.
(b) Prove that $P_{n,m} = (n-m)/(n+m)$.

Solution.

(a)

Step 1. Each ordering of n A's and m B's that ends with A corresponds to exactly one ordering of $n-1$ A's and m B's (delete the last letter), so the last vote is A's with probability $\binom{n+m-1}{n-1}/\binom{n+m}{n} = n/(n+m)$, and given this the first $n+m-1$ votes are a uniformly random ordering of $n-1$ A's and m B's. The same holds with the roles exchanged when the last vote is B's.

Step 2. Since $n > m$, A is ahead after the final vote, so A is ahead throughout iff A is ahead throughout the first $n+m-1$ votes. Hence for $m \geq 1$

$$P_{n,m} = \frac{n}{n+m}P_{n-1,m} + \frac{m}{n+m}P_{n,m-1},$$

with the conventions $P_{k,k} = 0$ for $k \geq 1$ (a count ending in a tie cannot keep A strictly ahead) and $P_{n,0} = 1$.

(b)

Step 1. We use induction on $n+m$. If $m=0$, every vote is A's and $P_{n,0}=1=n/n$. The formula also gives 0 when $n=m$, matching the convention for $P_{k,k}$.

Step 2. Let $n>m\geq 1$ and assume the formula for all pairs with smaller total (it covers $P_{n-1,m}$ also when $n-1=m$). By (a),

$$\begin{aligned} P_{n,m} &= \frac{n}{n+m}\cdot\frac{n-1-m}{n+m-1}+\frac{m}{n+m}\cdot\frac{n-m+1}{n+m-1} \\ &= \frac{(n-m)(n+m)-(n-m)}{(n+m)(n+m-1)} = \frac{n-m}{n+m}, \end{aligned}$$

where we used $n(n-m-1)+m(n-m+1)=n^2-m^2-(n-m)$. ∎

Lecture 4. Discrete Random Variables

4.1 Random variables and distribution functions

A **random variable** is a real function on the sample space. Its **(cumulative) distribution function** is $F(x) = P(X \le x)$. A random variable is **discrete** if it takes at most countably many values $x_1, x_2, \dots$; its **probability mass function** is $p(x) = P(X = x)$, with $\sum_i p(x_i) = 1$ and $F(a) = \sum_{x_i \le a} p(x_i)$, a step function.

Proposition 4.1 (Properties of F)

F is nondecreasing, right-continuous, $F(-\infty) = 0$ and $F(\infty) = 1$. Moreover $P(a < X \le b) = F(b) - F(a)$ and $P(X < b) = \lim_{x \uparrow b} F(x)$.

Proof. Monotonicity is Proposition 2.1(b). If $x_n \downarrow b$, the events $\{X \le x_n\}$ decrease to $\{X \le b\}$; if $x_n \uparrow \infty$ they increase to S; if $x_n \uparrow b$ the events $\{X \le x_n\}$ increase to $\{X < b\}$. Apply continuity (Proposition 2.3). ∎

Example 4.1 (Coupon collecting)

Each purchase gives one of N coupon types, equally likely and independently. Let T be the number of purchases needed to collect all types. With A_j the event that type j is missing after n purchases, inclusion–exclusion gives

$$P(T > n) = P\Big(\bigcup_j A_j\Big) = \sum_{i=1}^{N-1} (-1)^{i+1} \binom{N}{i} \Big(\frac{N-i}{N}\Big)^n,$$

and $P(T = n) = P(T > n-1) - P(T > n)$.

4.2 Expectation

Motivation. A random variable's long-run average is its expectation. It summarizes the distribution in one number and is the basis of fair prices, variance, and the laws of large numbers.

Definition (Expected value)

$E[X] = \sum_x x p(x)$, provided the sum converges absolutely.

It is the long-run average of repeated observations and the centre of gravity of the mass function. For the indicator I_A of an event, $E[I_A] = P(A)$.

Proposition 4.2

(a) $E[g(X)] = \sum_x g(x)p(x)$.
(b) $E[aX + b] = aE[X] + b$.
(c) On a countable sample space, $E[X] = \sum_{s \in S} X(s)P(\{s\})$, and hence $E[\sum_i X_i] = \sum_i E[X_i]$.

Proof. (a) Group the values x by $y = g(x)$: $\sum_y y P(g(X) = y) = \sum_y \sum_{x: g(x)=y} g(x)p(x) = \sum_x g(x)p(x)$.

(b) is a special case.

(c) Group outcomes s by the value $X(s)$; then linearity of the sum over s gives the second claim. ■

In general $E[g(X)] \neq g(E[X])$.

Example 4.2 (Which question first?)

A quiz contestant may answer question 1 or 2 first, and goes on to the other only if the first is answered correctly. Question i is answered correctly with probability p_i, independently, and is worth V_i. Answering 1 first gives expected winnings $p_1(1-p_2)V_1 + p_1p_2(V_1+V_2) = p_1V_1 + p_1p_2V_2$; the other order gives $p_2V_2 + p_1p_2V_1$. So question 1 should come first iff

$$\frac{p_1V_1}{1-p_1} > \frac{p_2V_2}{1-p_2}.$$

Example 4.3 (The friendship paradox)

In a network, let d_i be the number of friends of person i, and pick a random friendship (edge) and then one of its two ends at random. That person has D^* friends, and $P(D^* = k) = k\,P(D=k)/E[D]$, where D is the degree of a uniformly chosen person: a person with k friends is k times as likely to be reached. Hence

$$E[D^*] = \frac{E[D^2]}{E[D]} \geq E[D],$$

with equality only if all degrees are equal (by the variance inequality below). "Your friends have more friends than you", on average.

4.3 Variance

Definition (Variance)

With $\mu = E[X]$, $\operatorname{Var}(X) = E[(X-\mu)^2] = E[X^2] - \mu^2$. The standard deviation is $\sqrt{\operatorname{Var}(X)}$.

The second formula follows by expanding the square and using Proposition 4.2. As $\operatorname{Var}(X) \geq 0$, we get $E[X^2] \geq (E[X])^2$. Also $\operatorname{Var}(aX+b) = a^2 \operatorname{Var}(X)$. For a fair die, $E[X] = 7/2$, $E[X^2] = 91/6$, so $\operatorname{Var}(X) = 35/12$.

4.4 Bernoulli and binomial random variables

$X \sim \operatorname{Bin}(n,p)$ counts successes in n independent trials with success probability p:

$$p(k) = \binom{n}{k} p^k (1-p)^{n-k}, \qquad k = 0, \ldots, n.$$

Each specific sequence with k successes has probability $p^k(1-p)^{n-k}$ and there are $\binom{n}{k}$ such sequences. $\operatorname{Bin}(1,p)$ is the Bernoulli distribution.

Proposition 4.3

If $X \sim \operatorname{Bin}(n,p)$ then $E[X^k] = np\,E[(Y+1)^{k-1}]$ with $Y \sim \operatorname{Bin}(n-1,p)$. Hence $E[X] = np$ and $\operatorname{Var}(X) = np(1-p)$.

Proof. Using $i\binom{n}{i} = n\binom{n-1}{i-1}$,

$$E[X^k] = \sum_{i=1}^{n} i^{k-1}\, n\binom{n-1}{i-1} p^i (1-p)^{n-i} = np \sum_{j=0}^{n-1} (j+1)^{k-1} \binom{n-1}{j} p^j (1-p)^{n-1-j}.$$

$k=1$ gives np; $k=2$ gives $E[X^2] = np\{(n-1)p+1\}$, so $\mathrm{Var}(X) = np(1-p)$. ■

Since $p(k)/p(k-1) = \frac{(n-k+1)p}{k(1-p)}$, the pmf increases while $k < (n+1)p$ and then decreases: it is unimodal with mode $\lfloor (n+1)p \rfloor$. The ratio also gives a stable recursion for computing the cdf.

Example 4.4 (Chuck-a-luck)

A player bets 1 on a number; three dice are rolled and the player wins i if the number appears $i \geq 1$ times, and loses 1 otherwise. The number of appearances is $\mathrm{Bin}(3, 1/6)$, so the expected gain is

$$-\tfrac{125}{216} + 1 \cdot \tfrac{75}{216} + 2 \cdot \tfrac{15}{216} + 3 \cdot \tfrac{1}{216} = -\tfrac{17}{216} \approx -0.079.$$

Example 4.5 (k-out-of-n systems)

A system of n components, each working independently with probability p, works if at least half of them work. For which p is a 5-component system better than a 3-component one? $P_5 = \binom{5}{3}p^3q^2 + 5p^4q + p^5$ and $P_3 = 3p^2q + p^3$. One finds $P_5 - P_3 = 3p^2(p-1)^2(2p-1)$, so the larger system is better exactly when $p > \frac{1}{2}$.

4.5 The Poisson random variable

Motivation. Counts of rare events (typos, arrivals, defects) are approximately Poisson: a binomial with large n and small p converges to it. Its single parameter is both mean and variance.

$X \sim \mathrm{Poisson}(\lambda)$ if $p(k) = e^{-\lambda}\lambda^k/k!$, $k = 0, 1, \ldots$; then $E[X] = \mathrm{Var}(X) = \lambda$ (by the same shift trick: $E[X] = \lambda \sum_j e^{-\lambda}\lambda^j/j!$ and $E[X^2] = \lambda E[X+1]$).

Proposition 4.4 (Poisson approximation)

If $n \to \infty$ and $p \to 0$ with $np = \lambda$ fixed, then $\binom{n}{k}p^k(1-p)^{n-k} \to e^{-\lambda}\lambda^k/k!$.

Proof. $\binom{n}{k}p^k(1-p)^{n-k} = \dfrac{n(n-1)\cdots(n-k+1)}{n^k} \cdot \dfrac{\lambda^k}{k!} \cdot \dfrac{(1-\lambda/n)^n}{(1-\lambda/n)^k}$, and the three factors tend to 1, $\lambda^k/k!$ and $e^{-\lambda}$. ■

So the number of rare events among many trials (misprints on a page, people over 100, wrong numbers dialled) is approximately Poisson. The approximation remains good when the trials are only *weakly dependent* and success probabilities are small but unequal, with $\lambda = \sum p_i$ (the Poisson paradigm).

Example 4.6 (Poisson paradigm)

(a) Matching: the events "person i gets own hat" have probability $1/N$ and are weakly dependent, so the number of matches is approximately Poisson(1), as found exactly in Example 2.4.
(b) Birthdays: each of the $\binom{n}{2}$ pairs shares a birthday with probability $1/365$, and pair events

are weakly dependent, so $P(\text{no shared birthday}) \approx \exp\{-n(n-1)/730\}$. For $n = 23$ this is $e^{-253/365} \approx 0.500$; the exact value is 0.493.

Example 4.7 (Poisson process)

Events (earthquakes, arrivals) occur in time so that (i) counts in disjoint intervals are independent, (ii) in a short interval of length h, one event occurs with probability $\lambda h + o(h)$ and two or more with probability $o(h)$. Splitting $[0, t]$ into n pieces, the count $N(t)$ is approximately $\text{Bin}(n, \lambda t/n)$, and letting $n \to \infty$,

$$P(N(t) = k) = e^{-\lambda t} \frac{(\lambda t)^k}{k!}.$$

4.6 Other discrete distributions

Geometric. The number X of trials up to and including the first success: $p(k) = (1-p)^{k-1}p$, $k \geq 1$, and $P(X > k) = (1-p)^k$. With $q = 1 - p$, conditioning on the first trial gives $E[X] = p \cdot 1 + q(1 + E[X])$, so $E[X] = 1/p$; similarly $E[X^2] = p + q\,E[(1+X)^2]$ gives $E[X^2] = (2-p)/p^2$ and $\text{Var}(X) = q/p^2$.

Negative binomial. The number X of trials needed for r successes: the r-th success at trial k requires $r-1$ successes in the first $k-1$ trials, so

$$p(k) = \binom{k-1}{r-1} p^r (1-p)^{k-r}, \quad k \geq r, \qquad E[X] = \frac{r}{p}, \quad \text{Var}(X) = \frac{r(1-p)}{p^2}.$$

(X is a sum of r independent geometric waiting times, so Lecture 7 gives the moments at once.)

Example 4.8 (Banach match problem)

A smoker carries two boxes of N matches and each time picks a box at random. When a box is first found empty, what is the probability that the other contains exactly k matches? Call choosing the left box a "success". The left box is found empty with k matches on the right when success number $N+1$ occurs at trial $N + 1 + N - k$. So the probability is $2\binom{2N-k}{N}(\frac{1}{2})^{2N-k+1} = \binom{2N-k}{N}(\frac{1}{2})^{2N-k}$, the factor 2 accounting for the symmetric case.

Hypergeometric. Draw n balls without replacement from N, of which m are white; the number X of white balls drawn has

$$P(X = i) = \frac{\binom{m}{i}\binom{N-m}{n-i}}{\binom{N}{n}}, \qquad E[X] = \frac{nm}{N}, \qquad \text{Var}(X) = n\frac{m}{N}\Big(1 - \frac{m}{N}\Big)\frac{N-n}{N-1}.$$

With $p = m/N$ this is the binomial mean, and the variance carries the finite-population correction $(N-n)/(N-1)$. When N is large relative to n, X is approximately $\text{Bin}(n, m/N)$.

Example 4.9 (Capture–recapture)

To estimate an unknown population size N, catch and tag m animals, release them, and later catch n; suppose i are tagged. The probability of this, $P_i(N) = \binom{m}{i}\binom{N-m}{n-i}/\binom{N}{n}$, satisfies

$$\frac{P_i(N)}{P_i(N-1)} = \frac{(N-m)(N-n)}{N(N-m-n+i)} \geq 1 \iff N \leq \frac{mn}{i},$$

so the value of N that makes the observation most likely (the maximum likelihood estimate) is $\lfloor mn/i \rfloor$: the tagged fraction in the sample, i/n, is set equal to m/N.

Zeta (Zipf). $p(k) = C/k^{\alpha+1}$, $k \geq 1$, with $C = 1/\zeta(\alpha+1)$; it models heavy-tailed frequencies such as word counts.

4.7 Expected value of sums

By Proposition 4.2(c), $E[X_1+\cdots+X_n] = E[X_1]+\cdots+E[X_n]$ for *any* random variables, independent or not. Writing a count as a sum of indicators is the main trick of the course.

Example 4.10

(a) The sum of n fair dice has mean $7n/2$.
(b) If trial i succeeds with probability p_i, the expected number of successes is $\sum p_i$.
(c) The hypergeometric mean: with I_j the indicator that white ball j is drawn, $E[X] = \sum_{j=1}^{m} P(I_j = 1) = m \cdot n/N$ (Example 2.1(b)).

Example 4.11 (Variance of the number of successes)

With $X = \sum_i I_i$, $X^2 = \sum_i I_i + \sum_{i \neq j} I_i I_j$, so $E[X^2] = \sum_i p_i + \sum_{i \neq j} P(I_i = I_j = 1)$. For independent trials with common p, $E[X^2] = np + n(n-1)p^2$ and $\mathrm{Var}(X) = np(1-p)$. For the hypergeometric, $P(I_i = I_j = 1) = \frac{m(m-1)}{N(N-1)}$ for two specific draws $i \neq j$, which gives the variance stated above.

4.8 Selected exercises

Motivation. These are Ross's theoretical exercises for Chapter 4 (book numbering). They practise the two basic computational moves for discrete distributions, re-indexing a binomial sum and conditioning on an underlying count, and connect the Poisson, gamma, binomial and negative binomial distributions.

Exercise 4.10 (A reciprocal moment of the binomial)

Let $X \sim \mathrm{Bin}(n, p)$ with $0 < p \leq 1$. Show that

$$E\Big[\frac{1}{X+1}\Big] = \frac{1-(1-p)^{n+1}}{(n+1)p}.$$

Solution.

Step 1. For $0 \le i \le n$,

$$\frac{1}{i+1}\binom{n}{i} = \frac{n!}{(i+1)!\,(n-i)!} = \frac{1}{n+1}\binom{n+1}{i+1}.$$

Step 2. With $q = 1-p$ and Proposition 4.2(a),

$$\begin{aligned} E\Big[\frac{1}{X+1}\Big] &= \sum_{i=0}^{n} \frac{1}{i+1}\binom{n}{i} p^i q^{n-i} = \frac{1}{(n+1)p}\sum_{i=0}^{n}\binom{n+1}{i+1} p^{i+1} q^{n-i} \\ &= \frac{1}{(n+1)p}\sum_{j=1}^{n+1}\binom{n+1}{j} p^j q^{n+1-j} = \frac{(p+q)^{n+1} - q^{n+1}}{(n+1)p}, \end{aligned}$$

by the binomial theorem (Theorem 1.3) with the $j=0$ term removed. Since $p+q=1$ this is the claim. ∎

Exercise 4.25 (Thinning a Poisson count)

The number N of events in a fixed period is Poisson(λ), and each event is counted with probability p, independently of the other events and of N. Show that the number M of counted events is Poisson(λp). If $\lambda = 10$ uranium deposits are expected in an area and each is discovered with probability $1/50$, find the probability that at most one deposit is discovered.

Solution.

Step 1. Given $N = n$, M is the number of successes in n independent trials with success probability p, so $P(M = k \mid N = n) = \binom{n}{k} p^k q^{n-k}$ for $k \le n$ and 0 for $k > n$, where $q = 1-p$.

Step 2. By the law of total probability (Theorem 3.2, extended to the countable partition $\{N = n\}$ by countable additivity),

$$\begin{aligned} P(M=k) &= \sum_{n=k}^{\infty} e^{-\lambda}\frac{\lambda^n}{n!}\cdot\frac{n!}{k!\,(n-k)!}p^k q^{n-k} \\ &= e^{-\lambda}\frac{(\lambda p)^k}{k!}\sum_{j=0}^{\infty}\frac{(\lambda q)^j}{j!} = e^{-\lambda}\frac{(\lambda p)^k}{k!}e^{\lambda q} = e^{-\lambda p}\frac{(\lambda p)^k}{k!}. \end{aligned}$$

Step 3. Intuitively, N is approximately $\mathrm{Bin}(n, \lambda/n)$ for large n, and then M is approximately $\mathrm{Bin}(n, \lambda p/n)$, whose Poisson limit is Poisson(λp) by Proposition 4.4.

Step 4. Here $\lambda p = 0.2$, so $P(M \le 1) = e^{-0.2}(1 + 0.2) \approx 0.982$. ∎

Exercise 4.26 (Poisson tails as gamma integrals)

Prove that for $\lambda > 0$ and $n = 0, 1, 2, \dots$,

$$\sum_{i=0}^{n} e^{-\lambda}\frac{\lambda^i}{i!} = \frac{1}{n!}\int_{\lambda}^{\infty} e^{-x}x^n\,dx.$$

Solution.

Step 1. Put $I_n = \frac{1}{n!}\int_{\lambda}^{\infty} e^{-x}x^n\,dx$, which is finite. Then $I_0 = \int_{\lambda}^{\infty} e^{-x}\,dx = e^{-\lambda}$, which is the claim for $n = 0$.

Step 2. For $n \ge 1$ integrate by parts with $u = x^n/n!$ and $dv = e^{-x}dx$. Since $x^n e^{-x} \to 0$ as $x \to \infty$,

$$I_n = \Big[-e^{-x}\frac{x^n}{n!}\Big]_{\lambda}^{\infty} + \int_{\lambda}^{\infty} e^{-x}\frac{x^{n-1}}{(n-1)!}\,dx = e^{-\lambda}\frac{\lambda^n}{n!} + I_{n-1}.$$

Step 3. By induction on n, $I_n = \sum_{i=0}^{n} e^{-\lambda}\lambda^i/i!$. In words, if $X \sim \text{Poisson}(\lambda)$ and $Y \sim \text{Gamma}(n+1, 1)$, then $P(X \le n) = P(Y > \lambda)$: in a rate-one Poisson process, at most n events occur by time λ iff the $(n+1)$-st event occurs after λ. ■

Exercise 4.28 (Negative binomial versus binomial)

Let X be negative binomial with parameters (r, p) and $Y \sim \text{Bin}(n, p)$. Show that $P(X > n) = P(Y < r)$.

Solution.

Step 1. Consider one infinite sequence of independent trials with success probability p. Let X be the trial on which the r-th success occurs ($X = \infty$ if there is none) and Y the number of successes among the first n trials. Then X has the negative binomial distribution with parameters (r, p) (Section 4.6) and $Y \sim \text{Bin}(n, p)$, so it suffices to show $\{X > n\} = \{Y < r\}$ as events.

Step 2. If $Y \ge r$, the first n trials contain at least r successes, so the r-th success occurs at or before trial n and $X \le n$. Conversely, if $X \le n$, the first $X \le n$ trials already contain r successes, so $Y \ge r$. Hence $\{X \le n\} = \{Y \ge r\}$, and taking complements gives the claim. ■

Lecture 5. Continuous Random Variables

5.1 Densities

X is **continuous** if there is a nonnegative f (its **probability density**) with $P(X \in B) = \int_B f(x)\,dx$ for all (reasonable) $B \subset \mathbb{R}$. Then $\int_{-\infty}^{\infty} f = 1$, $P(a \le X \le b) = \int_a^b f$, $P(X = a) = 0$, and $F(a) = \int_{-\infty}^{a} f$, so $F' = f$ wherever f is continuous. Intuitively $P(a - \frac{\epsilon}{2} \le X \le a + \frac{\epsilon}{2}) \approx \epsilon f(a)$: $f(a)$ measures how likely X is to be *near* a.

Example 5.1

(a) If $f(x) = C(4x - 2x^2)$ on $(0, 2)$ and 0 elsewhere, then $1 = C\int_0^2 (4x - 2x^2)\,dx = 8C/3$, so $C = 3/8$; by symmetry about $x = 1$, $P(X > 1) = 1/2$.
(b) If $Y = 2X$ then $F_Y(a) = F_X(a/2)$ and $f_Y(a) = \frac{1}{2} f_X(a/2)$.

5.2 Expectation and variance

Motivation. For continuous variables sums become integrals. The tail-integral formula $E[X] = \int_0^\infty P(X > x)dx$ is often the quickest way to compute an expectation.

$E[X] = \int x f(x)\,dx$, and $\mathrm{Var}(X) = E[(X - \mu)^2] = E[X^2] - \mu^2$ as before.

Lemma 5.1 (Tail formula)

If $Y \ge 0$, then $E[Y] = \int_0^\infty P(Y > y)\,dy$.

Proof. $\int_0^\infty P(Y > y)\,dy = \int_0^\infty \int_y^\infty f_Y(x)\,dx\,dy = \int_0^\infty \left(\int_0^x dy\right) f_Y(x)\,dx = \int_0^\infty x f_Y(x)\,dx$. ∎

(The discrete analogue is $E[N] = \sum_{k \ge 1} P(N \ge k)$ for integer $N \ge 0$.)

Proposition 5.2

$E[g(X)] = \int g(x) f(x)\,dx$.

Proof. For $g \ge 0$, by the lemma, $E[g(X)] = \int_0^\infty P(g(X) > y)\,dy = \int_0^\infty \int_{x:g(x)>y} f(x)\,dx\,dy = \int \left(\int_0^{g(x)} dy\right) f(x)\,dx = \int g(x) f(x)\,dx$. In general split $g = g^+ - g^-$. ∎

Example 5.2 (Breaking a stick)

A stick of length 1 is broken at a uniform point U. What is the expected length of the piece containing a fixed point $p \in [0, 1]$? The length is $L_p(U) = 1 - U$ if $U < p$ and U if $U > p$, so

$$E[L_p(U)] = \int_0^p (1 - u)\,du + \int_p^1 u\,du = \frac{1}{2} + p(1 - p).$$

It is at least $\frac{1}{2}$ and maximal for $p = \frac{1}{2}$: the piece containing p tends to be the longer one, because longer pieces are more likely to contain p (compare Example 4.3).

Example 5.3 (When to leave)

Travel time X has density f and cdf F. Arriving s minutes early costs cs; arriving s minutes late costs ks. Leaving t minutes before the appointment costs, in expectation,

$$E[C_t(X)] = \int_0^t c(t-x)f(x)\,dx + \int_t^\infty k(x-t)f(x)\,dx, \qquad \frac{d}{dt}E[C_t(X)] = cF(t) - k\{1-F(t)\}.$$

The optimum is the quantile $F(t^*) = k/(k+c)$; for $k = c$ it is the median. The same argument shows that $E|X-a|$ is minimized at the median of X.

5.3 The uniform distribution

$X \sim U(\alpha, \beta)$ has density $1/(\beta-\alpha)$ on (α, β). Then $E[X] = (\alpha+\beta)/2$ and $\mathrm{Var}(X) = (\beta-\alpha)^2/12$.

Example 5.4 (Bertrand's paradox)

What is the probability that a "random chord" of a circle of radius r is longer than the side $\sqrt{3}r$ of the inscribed equilateral triangle? A chord at distance D from the centre has length $2\sqrt{r^2-D^2} > \sqrt{3}r$ iff $D < r/2$.
(i) If $D \sim U(0, r)$ (pick a random point on a radius and draw the perpendicular chord), the answer is $1/2$.
(ii) If one endpoint is fixed and the angle θ between the chord and the tangent there is $U(0, \pi)$ (equivalently, the other endpoint is uniform on the circle), the chord has length $2r\sin\theta$, which exceeds $\sqrt{3}r$ iff $\theta \in (\pi/3, 2\pi/3)$: probability $1/3$. The answers differ because "random chord" has not been defined: different experiments give different distributions.

5.4 Normal random variables

$X \sim N(\mu, \sigma^2)$ has density

$$f(x) = \frac{1}{\sqrt{2\pi}\,\sigma} e^{-(x-\mu)^2/(2\sigma^2)}, \qquad -\infty < x < \infty.$$

That it integrates to 1 follows from $I^2 = \iint e^{-(x^2+y^2)/2}\,dx\,dy = \int_0^{2\pi}\int_0^\infty e^{-r^2/2} r\,dr\,d\theta = 2\pi$ for $I = \int e^{-x^2/2}\,dx$. If $X \sim N(\mu, \sigma^2)$ then $aX + b \sim N(a\mu + b, a^2\sigma^2)$, so $Z = (X-\mu)/\sigma$ is **standard normal**, with cdf Φ and $\Phi(-x) = 1 - \Phi(x)$. For Z, $E[Z] = 0$ by symmetry and $E[Z^2] = 1$ by integration by parts; hence $E[X] = \mu$, $\mathrm{Var}(X) = \sigma^2$, and

$$P(X \le a) = \Phi\Big(\frac{a-\mu}{\sigma}\Big).$$

Useful values: $P(|Z| < 1) \approx 0.683$, $P(|Z| < 2) \approx 0.954$, $P(|Z| < 3) \approx 0.997$; $\Phi(1.645) = 0.95$, $\Phi(1.96) = 0.975$. For large x, $1 - \Phi(x) \approx e^{-x^2/2}/(x\sqrt{2\pi})$.

Example 5.5 (Detecting a signal)

A bit is sent as $+2$ (for 1) or -2 (for 0); the receiver gets $R = \pm 2 + N$ with $N \sim N(0, 1)$ and decodes 1 iff $R \ge 0.5$. The two error probabilities are $P(N < -1.5) = 1 - \Phi(1.5) \approx 0.067$ and $P(N \ge 2.5) = 1 - \Phi(2.5) \approx 0.006$.

Theorem 5.3 (De Moivre–Laplace)

If $S_n \sim \mathrm{Bin}(n,p)$, then for $a<b$, $P\Big(a \le \dfrac{S_n - np}{\sqrt{np(1-p)}} \le b\Big) \to \Phi(b)-\Phi(a)$ as $n\to\infty$.

This is a special case of the central limit theorem (Lecture 8). The approximation is good when $np(1-p)\ge 10$. Because S_n is integer-valued we use the **continuity correction**: $P(S_n = i) = P(i-\tfrac12 < S_n < i+\tfrac12)$ before standardizing. (For small p and np moderate, use the Poisson approximation instead.)

Example 5.6

(a) For 40 fair flips, $P(S=20) \approx \Phi(0.5/\sqrt{10}) - \Phi(-0.5/\sqrt{10}) \approx 0.1256$; the exact value is $\binom{40}{20}2^{-40} \approx 0.1254$.
(b) A college admits 450 students, each enrolling with probability 0.3 independently, and has room for 150. $S \sim \mathrm{Bin}(450, 0.3)$ has mean 135 and variance 94.5, so $P(S \ge 151) = P(S \ge 150.5) \approx 1 - \Phi(15.5/\sqrt{94.5}) = 1-\Phi(1.59) \approx 0.055$ (exact: 0.057).

5.5 Exponential random variables

$X \sim \mathrm{Exp}(\lambda)$ has $f(x) = \lambda e^{-\lambda x}$ and $P(X>x) = e^{-\lambda x}$ for $x \ge 0$; $E[X] = 1/\lambda$, $\mathrm{Var}(X) = 1/\lambda^2$. In a Poisson process with rate λ (Example 4.7), the time to the first event exceeds t iff $N(t)=0$, which has probability $e^{-\lambda t}$, so waiting times are exponential.

Proposition 5.4 (Memorylessness)

$P(X > s+t \mid X > t) = P(X>s)$ for all $s,t\ge 0$ holds for $X\sim\mathrm{Exp}(\lambda)$, and the exponential distributions are the only continuous distributions on $[0,\infty)$ with this property.

Proof. For the exponential, $e^{-\lambda(s+t)}/e^{-\lambda t} = e^{-\lambda s}$. Conversely, $\bar F(x) = P(X>x)$ satisfies $\bar F(s+t) = \bar F(s)\bar F(t)$. Then $\bar F(m/n) = \bar F(1/n)^m$ and $\bar F(1) = \bar F(1/n)^n$, so $\bar F(x) = \bar F(1)^x$ for rational x, and by right continuity for all $x\ge 0$; i.e. $\bar F(x) = e^{-\lambda x}$ with $\lambda = -\log \bar F(1)$. ∎

Example 5.7 (Two clerks)

A customer enters a post office where clerks are serving Ms Jones and Mr Brown, and will be served as soon as either finishes. Service times are independent exponentials with the same rate. What is the probability that the customer is the last of the three to leave? When the first of Jones and Brown leaves, the other's remaining service time is again exponential with the same rate, by memorylessness, so the newcomer and the remaining customer are in a symmetric position: the probability is $1/2$.

Hazard rate. For a positive continuous lifetime X with cdf F, the **hazard (failure) rate** is $\lambda(t) = f(t)/\bar F(t)$; $\lambda(t)\,dt$ is the probability that an item of age t fails in the next dt. Since $\lambda(t) = -\frac{d}{dt}\log \bar F(t)$,

$$\bar F(t) = \exp\Big\{-\int_0^t \lambda(s)\,ds\Big\},$$

so λ determines the distribution. A constant hazard characterizes the exponential; increasing hazards describe ageing.

Example 5.8

If at every age a smoker's death rate is twice a non-smoker's, the probability that a smoker of age A survives to age B is $\exp\{-2\int_A^B \lambda_N\}$, the *square* of a non-smoker's survival probability. If the latter is 0.76 from age 50 to 60, the smoker's is 0.578.

5.6 Other continuous distributions

Gamma. $X \sim \text{Gamma}(\alpha, \lambda)$ has $f(x) = \lambda e^{-\lambda x}(\lambda x)^{\alpha-1}/\Gamma(\alpha)$, $x > 0$, where $\Gamma(\alpha) = \int_0^\infty e^{-y} y^{\alpha-1}\, dy$ satisfies $\Gamma(\alpha) = (\alpha-1)\Gamma(\alpha-1)$ (integration by parts), $\Gamma(n) = (n-1)!$, and $\Gamma(\frac{1}{2}) = \sqrt{\pi}$. Then $E[X] = \alpha/\lambda$ and $\text{Var}(X) = \alpha/\lambda^2$. The time T_n of the n-th event of a Poisson process is $\text{Gamma}(n, \lambda)$, because

$$P(T_n \le t) = P(N(t) \ge n) = \sum_{j \ge n} e^{-\lambda t} \frac{(\lambda t)^j}{j!},$$

and differentiating in t the sum telescopes to $\lambda e^{-\lambda t}(\lambda t)^{n-1}/(n-1)!$. The **chi-squared** distribution with n degrees of freedom is $\text{Gamma}(n/2, 1/2)$.

Weibull. $F(x) = 1 - \exp\{-((x-\nu)/\alpha)^\beta\}$ for $x > \nu$; it models lifetimes (weakest-link failures) and has hazard proportional to $(x-\nu)^{\beta-1}$.

Cauchy. $f(x) = \frac{1}{\pi}\frac{1}{1+(x-\theta)^2}$. A flashlight at distance 1 from a wall is spun so that its angle θ to the perpendicular is $U(-\pi/2, \pi/2)$; the point lit on the wall is $X = \tan\theta$, and $P(X \le x) = P(\theta \le \arctan x) = \frac{1}{2} + \frac{1}{\pi}\arctan x$, the standard Cauchy. Its mean does not exist, since $\int |x|/(1+x^2)\, dx = \infty$.

Beta. $f(x) = x^{a-1}(1-x)^{b-1}/B(a,b)$ on $(0,1)$, with $B(a,b) = \Gamma(a)\Gamma(b)/\Gamma(a+b)$, $E[X] = a/(a+b)$ and $\text{Var}(X) = ab/\{(a+b)^2(a+b+1)\}$. It models a random proportion; $a = b = 1$ is uniform.

Pareto. If $X \sim \text{Exp}(\lambda)$ and $a > 0$, then $Y = ae^X$ has $P(Y > y) = (a/y)^\lambda$ for $y \ge a$. Given $Y > y_0$, Y is again Pareto with a replaced by y_0: incomes above a threshold look like rescaled incomes.

5.7 The distribution of a function of a random variable

Motivation. If $Y = g(X)$, we usually want the density of Y. Computing the cdf first and differentiating always works; for monotone g it gives a formula with $|dg^{-1}/dy|$.

The general method is to compute the cdf of $Y = g(X)$ and differentiate.

Theorem 5.5

If g is strictly monotone and differentiable on the range of X, then $Y = g(X)$ has density

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|$$

for y in the range of g, and 0 otherwise.

Proof. For increasing g, $F_Y(y) = P(X \le g^{-1}(y)) = F_X(g^{-1}(y))$; differentiate. For decreasing g, $F_Y(y) = 1 - F_X(g^{-1}(y))$, and the derivative of g^{-1} is negative. ∎

Example 5.9

(a) $Y = X^2$ (not monotone): for $y > 0$, $F_Y(y) = F_X(\sqrt{y}) - F_X(-\sqrt{y})$, so $f_Y(y) = \frac{1}{2\sqrt{y}}\{f_X(\sqrt{y}) + f_X(-\sqrt{y})\}$. With $X = Z$ standard normal this is $\frac{1}{\sqrt{2\pi y}}e^{-y/2}$, the chi-squared density with one degree of freedom (and shows $\Gamma(\frac{1}{2}) = \sqrt{\pi}$).
(b) $|X|$ has density $f_X(y) + f_X(-y)$ for $y > 0$.
(c) Lognormal: if $X \sim N(\mu, \sigma^2)$, $Y = e^X$ has $f_Y(y) = \frac{1}{\sqrt{2\pi}\sigma y}\exp\{-(\log y - \mu)^2/(2\sigma^2)\}$, $y > 0$. It models prices, where daily ratios multiply.
(d) If F is continuous and strictly increasing, then $F(X) \sim U(0,1)$ and $F^{-1}(U)$ has cdf F for $U \sim U(0,1)$; this is how random variables are simulated.

5.8 Selected exercises

Motivation. These are Ross's theoretical exercises for Chapter 5 (book numbering). They extend the tail formula to higher moments, give a sharp variance bound for bounded variables, derive the normal moments by integration by parts, and prove the probability integral transform without assuming that F is strictly increasing.

Exercise 5.5 (Moments from the tail)

Use Lemma 5.1 to show that for a nonnegative random variable X and $n \geq 1$,

$$E[X^n] = \int_0^\infty nx^{n-1}P(X > x)\,dx.$$

Deduce $E[X^2]$ for $X \sim \text{Exp}(\lambda)$.

Solution.

Step 1. $Y = X^n$ is nonnegative, so Lemma 5.1 gives $E[X^n] = \int_0^\infty P(X^n > t)\,dt$.

Step 2. Substitute $t = x^n$, a strictly increasing C^1 bijection of $[0,\infty)$ onto itself with $dt = nx^{n-1}dx$. Since $X \geq 0$ and $x \geq 0$, $X^n > x^n$ iff $X > x$. Hence $E[X^n] = \int_0^\infty nx^{n-1}P(X > x)\,dx$.

Step 3. For $X \sim \text{Exp}(\lambda)$, $P(X > x) = e^{-\lambda x}$ and

$$E[X^2] = \int_0^\infty 2xe^{-\lambda x}\,dx = \frac{2}{\lambda^2},$$

so $\text{Var}(X) = 2/\lambda^2 - 1/\lambda^2 = 1/\lambda^2$, as stated in Section 5.5. ■

Exercise 5.8 (Variance of a bounded variable)

Let $P(0 \leq X \leq c) = 1$ for some $c > 0$. Show that $\text{Var}(X) \leq c^2/4$.

Solution.

Step 1. On $\{0 \leq X \leq c\}$ we have $X^2 \leq cX$, so $E[X^2] \leq cE[X]$.

Step 2. Put $\alpha = E[X]/c \in [0,1]$. Then

$$\text{Var}(X) = E[X^2] - (E[X])^2 \leq cE[X] - (E[X])^2 = c^2\alpha(1-\alpha) \leq \frac{c^2}{4},$$

since $\alpha(1-\alpha) = \frac{1}{4} - (\alpha - \frac{1}{2})^2 \leq \frac{1}{4}$.

Step 3. The bound is attained when $P(X = 0) = P(X = c) = \frac{1}{2}$: then $E[X] = c/2$, $E[X^2] = c^2/2$ and $\mathrm{Var}(X) = c^2/4$. ■

Exercise 5.11 (Stein's identity and normal moments)

Let $Z \sim N(0,1)$ with density ϕ, and let g be differentiable with $E|g'(Z)| < \infty$, $E|Zg(Z)| < \infty$ and $g(x)\phi(x) \to 0$ as $|x| \to \infty$.
(a) Show that $E[g'(Z)] = E[Zg(Z)]$.
(b) Show that $E[Z^{n+1}] = nE[Z^{n-1}]$ for $n \geq 1$.
(c) Find $E[Z^4]$.

Solution.

(a)

Step 1. $\phi(x) = e^{-x^2/2}/\sqrt{2\pi}$ satisfies $\phi'(x) = -x\phi(x)$.

Step 2. For $A, B > 0$, integration by parts gives

$$\int_{-A}^{B} g'(x)\phi(x)\,dx = g(B)\phi(B) - g(-A)\phi(-A) + \int_{-A}^{B} xg(x)\phi(x)\,dx.$$

Step 3. Let $A, B \to \infty$. The boundary terms tend to 0 by assumption, and both integrals converge to the integrals over $\mathbb{R}$ because $g'\phi$ and $xg\phi$ are integrable. By Proposition 5.2 this is $E[g'(Z)] = E[Zg(Z)]$.

(b) Take $g(x) = x^n$. Then $x^n\phi(x) \to 0$, and every moment of Z is finite because $|x|^k\phi(x) \leq C_k e^{-x^2/4}$, which is integrable. So (a) gives $E[Z^{n+1}] = E[nZ^{n-1}]$.

(c) With $E[Z^0] = 1$: $E[Z^2] = 1 \cdot E[Z^0] = 1$ and $E[Z^4] = 3E[Z^2] = 3$. In general $E[Z^{2k}] = (2k-1)(2k-3)\cdots 1$, and the odd moments vanish since $E[Z] = 0$. ■

Exercise 5.29 (The probability integral transform)

Let X be a continuous random variable with distribution function F. Show that $Y = F(X)$ is uniformly distributed on $(0,1)$. (Example 5.9(d) assumed F strictly increasing; here F may be constant on intervals.)

Solution.

Step 1. F is continuous (it has a density) and nondecreasing with $F(-\infty) = 0$, $F(\infty) = 1$ (Proposition 4.1). Since $0 \leq Y \leq 1$, it suffices to show $P(Y \leq y) = y$ for $0 < y < 1$, and $P(Y \leq 0) = 0$.

Step 2. Fix $y \in (0,1)$. The level set $L = \{x : F(x) = y\}$ is nonempty by the intermediate value theorem, closed by continuity, and bounded above because $F(x) \to 1 > y$. Let $q = \max L$, so $F(q) = y$.

Step 3. We claim $\{F(X) \leq y\} = \{X \leq q\}$. If $X \leq q$, then $F(X) \leq F(q) = y$ by monotonicity. If $X > q$, then $F(X) \geq F(q) = y$, and $F(X) \neq y$ because $X \notin L$ (as $X > \max L$); so $F(X) > y$.

Step 4. Hence $P(Y \leq y) = P(X \leq q) = F(q) = y$.

Step 5. If $F(x) > 0$ for all x, then $P(Y \leq 0) = P(F(X) = 0) = 0$. Otherwise $\{x : F(x) = 0\}$ is nonempty, closed and bounded above; with a its maximum, $\{F(X) = 0\} = \{X \leq a\}$ as in Step 3, so $P(Y \leq 0) = F(a) = 0$. ■

Lecture 6. Jointly Distributed Random Variables

6.1 Joint distribution functions

The **joint cdf** of X, Y is $F(a,b) = P(X \le a, Y \le b)$; the marginals are $F_X(a) = F(a,\infty)$ and $F_Y(b) = F(\infty, b)$, and

$$P(a_1 < X \le a_2,\ b_1 < Y \le b_2) = F(a_2,b_2) - F(a_1,b_2) - F(a_2,b_1) + F(a_1,b_1).$$

For discrete variables the joint pmf is $p(x,y) = P(X = x, Y = y)$, and $p_X(x) = \sum_y p(x,y)$. X and Y are **jointly continuous** with joint density f if $P((X,Y) \in C) = \iint_C f(x,y)\,dx\,dy$; then $f = \partial^2 F/\partial a\,\partial b$, and $f_X(x) = \int f(x,y)\,dy$. All of this extends to n variables.

Example 6.1

Let $f(x,y) = 2e^{-x}e^{-2y}$ for $x, y > 0$. Then $P(X > 1, Y < 1) = \int_0^1 \int_1^\infty 2e^{-x}e^{-2y}\,dx\,dy = e^{-1}(1 - e^{-2}) \approx 0.318$, and

$$P(X < Y) = \int_0^\infty \int_0^y 2e^{-x}e^{-2y}\,dx\,dy = \int_0^\infty 2e^{-2y}(1 - e^{-y})\,dy = 1 - \tfrac{2}{3} = \tfrac{1}{3}.$$

Example 6.2 (A random point in a disc)

(X,Y) is uniform on the disc of radius R: $f = 1/(\pi R^2)$ there. The marginal is $f_X(x) = \frac{2\sqrt{R^2 - x^2}}{\pi R^2}$ for $|x| \le R$, which is not uniform. The distance D to the centre has $P(D \le a) = \pi a^2/(\pi R^2) = a^2/R^2$, so $f_D(a) = 2a/R^2$ and $E[D] = 2R/3$.

Example 6.3 (Multinomial distribution)

In n independent trials, each resulting in outcome i with probability p_i ($\sum_1^r p_i = 1$), the counts $X_1, \dots, X_r$ satisfy

$$P(X_1 = n_1, \dots, X_r = n_r) = \frac{n!}{n_1! \cdots n_r!} p_1^{n_1} \cdots p_r^{n_r}, \qquad \sum n_i = n,$$

since each sequence with these counts has probability $\prod p_i^{n_i}$ and there are $\binom{n}{n_1,\dots,n_r}$ such sequences (Lecture 1). Each X_i is marginally $\mathrm{Bin}(n, p_i)$.

6.2 Independent random variables

Motivation. Independence means the joint distribution factors into marginals. It is what makes joint probabilities computable and underlies sums, products and the limit theorems.

X and Y are **independent** if $P(X \in A, Y \in B) = P(X \in A)P(Y \in B)$ for all A, B; equivalently $F(a,b) = F_X(a)F_Y(b)$, or $p(x,y) = p_X(x)p_Y(y)$, or $f(x,y) = f_X(x)f_Y(y)$.

Proposition 6.1 (Factorization)

Jointly continuous (or discrete) X, Y are independent iff the joint density (pmf) factors as $f(x,y) = h(x)g(y)$ for all x, y.

Proof. If $f = hg$, put $C_1 = \int h$, $C_2 = \int g$; then $C_1C_2 = \iint f = 1$, $f_X = hC_2$ and $f_Y = gC_1$, so $f_X f_Y = hg\,C_1C_2 = f$. ∎

The factorization must hold on the whole plane: $f = 24xy$ on $\{x, y > 0, x + y < 1\}$ looks like a product but is not, because the support is not a rectangle.

Example 6.4 (Thinning a Poisson count)

The number N of customers on a day is Poisson(λ), and each is independently a man with probability p. Let X and Y be the numbers of men and women. Then

$$P(X = i, Y = j) = e^{-\lambda}\frac{\lambda^{i+j}}{(i+j)!}\binom{i+j}{i}p^i q^j = e^{-\lambda p}\frac{(\lambda p)^i}{i!}\cdot e^{-\lambda q}\frac{(\lambda q)^j}{j!}:$$

X and Y are *independent* Poisson variables with means λp and λq. (With a fixed number n of customers, they would be strongly dependent.)

Example 6.5 (Meeting)

Two people arrive independently and uniformly between 12:15 and 12:45 and each waits 10 minutes. Measuring in minutes past 12:15, they meet iff $|X - Y| \le 10$, so $P = 1 - (20/30)^2 = 5/9$ (the two corner triangles of the square have total area 20^2).

Example 6.6 (Buffon's needle)

A needle of length L is dropped on a floor ruled with parallel lines $D \ge L$ apart. Let $X \sim U(0, D/2)$ be the distance from the needle's centre to the nearest line and $\theta \sim U(0, \pi/2)$ the acute angle with the lines, independent. The needle crosses a line iff $X < \frac{L}{2}\cos\theta$, so

$$P(\text{cross}) = \frac{4}{\pi D}\int_0^{\pi/2}\int_0^{(L/2)\cos\theta} dx\,d\theta = \frac{2L}{\pi D}.$$

Dropping many needles thus estimates π.

Example 6.7 (Characterizing the normal)

Darts land at (X, Y) with X, Y independent with a common differentiable density f, and the joint density depends only on the distance from the target: $f(x)f(y) = g(x^2 + y^2)$. Differentiating in x gives $f'(x)f(y) = 2xg'(x^2 + y^2)$, so $\frac{f'(x)}{2xf(x)} = \frac{g'(x^2+y^2)}{g(x^2+y^2)}$ for all x, y. The left side does not depend on y, and the right side then shows it does not depend on x; hence $f'(x)/f(x) = cx$, $f(x) = ke^{cx^2/2}$ with $c < 0$: X is normal with mean 0.

6.3 Sums of independent random variables

Motivation. The distribution of a sum of independent variables is the convolution of their distributions. Several families are closed under sums (normal, Poisson, gamma), which explains their importance.

If X, Y are independent with densities f_X, f_Y, then

$$F_{X+Y}(a) = \int F_X(a - y)f_Y(y)\,dy, \qquad f_{X+Y}(a) = \int f_X(a - y)f_Y(y)\,dy,$$

the **convolution** of f_X and f_Y.

Example 6.8 (Sums of uniforms)

For $X, Y \sim U(0,1)$ independent, $f_{X+Y}(a) = a$ on $[0,1]$ and $2-a$ on $[1,2]$ (triangular). More generally, if S_n is a sum of n independent uniforms, induction on n gives $P(S_n \le x) = x^n/n!$ for $0 \le x \le 1$. Let $N = \min\{n : S_n > 1\}$. Then $P(N > n) = P(S_n \le 1) = 1/n!$, so

$$E[N] = \sum_{n \ge 0} P(N > n) = \sum_{n \ge 0} \frac{1}{n!} = e.$$

Proposition 6.2

For independent variables:
(a) $\text{Gamma}(s,\lambda) + \text{Gamma}(t,\lambda) \sim \text{Gamma}(s+t,\lambda)$; in particular a sum of n independent $\text{Exp}(\lambda)$ is $\text{Gamma}(n,\lambda)$ and $Z_1^2 + \cdots + Z_n^2 \sim \chi^2_n$.
(b) $\sum N(\mu_i, \sigma_i^2) \sim N(\sum \mu_i, \sum \sigma_i^2)$.
(c) $\text{Poisson}(\lambda_1) + \text{Poisson}(\lambda_2) \sim \text{Poisson}(\lambda_1 + \lambda_2)$.
(d) $\text{Bin}(n,p) + \text{Bin}(m,p) \sim \text{Bin}(n+m,p)$.

Proof. (a) The convolution is $\frac{\lambda e^{-\lambda a}}{\Gamma(s)\Gamma(t)} \lambda^{s+t-1} \int_0^a y^{t-1}(a-y)^{s-1}\,dy$; substituting $y = ax$ shows it is $Ca^{s+t-1}e^{-\lambda a}$, a gamma density, and C is then forced.

(b) It suffices to treat $N(0,\sigma^2)+N(0,1)$, a direct completion of the square; Lecture 7 gives a one-line proof with moment generating functions.

(c) $P(X+Y=n) = \sum_k e^{-\lambda_1}\frac{\lambda_1^k}{k!} e^{-\lambda_2}\frac{\lambda_2^{n-k}}{(n-k)!} = e^{-(\lambda_1+\lambda_2)}\frac{(\lambda_1+\lambda_2)^n}{n!}$ by the binomial theorem.

(d) Count successes in $n+m$ trials, or use Vandermonde. ∎

Example 6.9

A team plays 26 games against class A teams, winning each with probability 0.4, and 18 against class B, winning each with probability 0.7, all independently. Its number of wins has mean $26(0.4) + 18(0.7) = 23$ and variance $26(0.24) + 18(0.21) = 10.02$; it is not binomial, but the normal approximation applies: $P(\text{wins} \ge 25) \approx 1 - \Phi(1.5/\sqrt{10.02}) \approx 0.32$.

6.4 Conditional distributions

For discrete variables, $p_{X|Y}(x \mid y) = p(x,y)/p_Y(y)$ when $p_Y(y) > 0$. For jointly continuous ones, the conditional density is

$$f_{X|Y}(x \mid y) = \frac{f(x,y)}{f_Y(y)}, \qquad P(X \in A \mid Y = y) = \int_A f_{X|Y}(x \mid y)\,dx,$$

which is motivated by $P(x < X < x+dx \mid y < Y < y+dy) \approx f(x,y)\,dx\,dy/(f_Y(y)\,dy)$. Independence means the conditional law does not depend on y.

Example 6.10

(a) If X, Y are independent Poisson with means λ_1, λ_2, then given $X + Y = n$, $X \sim \text{Bin}(n, \lambda_1/(\lambda_1+\lambda_2))$. Likewise, given $X + Y = n$ for independent $\text{Bin}(n_1,p)$ and $\text{Bin}(n_2,p)$, X is hypergeometric, free of p.

(b) In a multinomial, given $X_r = n_r$, the remaining counts are multinomial with $n - n_r$ trials and probabilities $p_i/(1-p_r)$.
(c) If $f(x,y) = e^{-x/y}e^{-y}/y$ for $x, y > 0$, then $f_{X|Y}(x \mid y) = \frac{1}{y}e^{-x/y}$: given $Y = y$, X is exponential with mean y, and $P(X > 1 \mid Y = y) = e^{-1/y}$.

Example 6.11 (The bivariate normal)

(X, Y) is bivariate normal with parameters $\mu_x, \mu_y, \sigma_x, \sigma_y, \rho$ ($|\rho| < 1$) if

$$f(x,y) = \frac{1}{2\pi\sigma_x\sigma_y\sqrt{1-\rho^2}} \exp\Big\{-\frac{1}{2(1-\rho^2)}\Big[u^2 - 2\rho uv + v^2\Big]\Big\}, \quad u = \frac{x-\mu_x}{\sigma_x},\ v = \frac{y-\mu_y}{\sigma_y}.$$

Completing the square in u shows that, given $Y = y$, X is normal with

$$E[X \mid Y = y] = \mu_x + \rho\frac{\sigma_x}{\sigma_y}(y - \mu_y), \qquad \mathrm{Var}(X \mid Y = y) = \sigma_x^2(1-\rho^2),$$

and integrating out x shows $Y \sim N(\mu_y, \sigma_y^2)$. X and Y are independent iff $\rho = 0$ (Lecture 7 shows ρ is the correlation).

Example 6.12 (Mixed variables: beta–binomial)

Let $P \sim U(0,1)$ and, given $P = p$, let $N \sim \mathrm{Bin}(n,p)$. Then $P(N = k) = \int_0^1 \binom{n}{k} p^k(1-p)^{n-k}\,dp = \binom{n}{k}B(k+1, n-k+1) = \frac{1}{n+1}$: the number of successes is uniform on $\{0, \dots, n\}$. Given $N = k$, the density of P is proportional to $p^k(1-p)^{n-k}$: $P \mid N = k \sim \mathrm{Beta}(k+1, n-k+1)$. Compare Laplace's rule of succession (Example 3.12).

6.5 Order statistics

Let $X_1, \dots, X_n$ be i.i.d. with density f and cdf F, and $X_{(1)} \le \cdots \le X_{(n)}$ their ordered values. Each of the $n!$ assignments of the X_i to the ordered values is equally likely, so

$$f_{X_{(1)},\dots,X_{(n)}}(x_1, \dots, x_n) = n!\, f(x_1)\cdots f(x_n), \qquad x_1 < \cdots < x_n.$$

For one order statistic: $X_{(j)}$ is near x when $j-1$ observations are below x, one is near x, and $n-j$ are above, so

$$f_{X_{(j)}}(x) = \frac{n!}{(j-1)!\,(n-j)!} F(x)^{j-1}(1-F(x))^{n-j} f(x).$$

Also $P(X_{(j)} \le x) = \sum_{k=j}^{n} \binom{n}{k} F(x)^k(1-F(x))^{n-k}$ (at least j observations below x). For uniforms, $X_{(j)} \sim \mathrm{Beta}(j, n-j+1)$, so $E[X_{(j)}] = j/(n+1)$.

Example 6.13

(a) Three people are placed independently and uniformly on a road of length 1. The probability that no two are within distance $d \le \frac{1}{2}$ is

$$3! \int_{x_1 < x_2 - d,\ x_2 < x_3 - d} dx_1\, dx_2\, dx_3 = (1-2d)^3,$$

by substituting $y_1 = x_1$, $y_2 = x_2 - d$, $y_3 = x_3 - 2d$, which maps the region onto the ordered simplex in $[0, 1-2d]^3$.

(b) The sample median of $2n+1$ uniforms has density $\frac{(2n+1)!}{n!\,n!}x^n(1-x)^n$, and the range $R = X_{(n)} - X_{(1)}$ has $P(R \le a) = n\int f(x)\{F(x+a)-F(x)\}^{n-1}\,dx$; for uniforms, $f_R(a) = n(n-1)a^{n-2}(1-a)$.

6.6 Functions of random variables: the Jacobian method

Motivation. Changing variables in a joint density works like change of variables in a double integral: the Jacobian determinant accounts for the stretching of area. This yields, for instance, the polar method for normals.

Theorem 6.3

Let (X_1, X_2) have density f, and let $y_1 = g_1(x_1,x_2)$, $y_2 = g_2(x_1,x_2)$ be a one-to-one differentiable map with inverse $x_i = h_i(y_1,y_2)$ and nonzero Jacobian $J(x_1,x_2) = \det(\partial g_i/\partial x_j)$. Then $Y_i = g_i(X_1,X_2)$ have joint density

$$f_{Y_1,Y_2}(y_1,y_2) = f\big(h_1(y),h_2(y)\big)\,|J(h_1(y),h_2(y))|^{-1}.$$

This is the change of variables formula for double integrals; it holds in n dimensions as well.

Example 6.14 (Polar coordinates of normals)

Let X, Y be independent $N(0,1)$, and $R = \sqrt{X^2+Y^2}$, $\Theta = \arctan(Y/X) \in [0, 2\pi)$. The Jacobian of $(x,y) \mapsto (r,\theta)$ is $1/r$, so

$$f_{R,\Theta}(r,\theta) = \frac{1}{2\pi}e^{-r^2/2}\,r = \frac{1}{2\pi}\cdot re^{-r^2/2}:$$

R and Θ are independent, Θ is uniform, and $R^2 \sim \mathrm{Exp}(1/2) = \chi^2_2$. Reversing this gives the Box–Muller method: $X = \sqrt{-2\log U_1}\cos(2\pi U_2)$ and $Y = \sqrt{-2\log U_1}\sin(2\pi U_2)$ are independent standard normals.

Example 6.15 (Gamma and beta)

Let $X \sim \mathrm{Gamma}(\alpha,\lambda)$ and $Y \sim \mathrm{Gamma}(\beta,\lambda)$ be independent, $U = X+Y$, $V = X/(X+Y)$. Then $x = uv$, $y = u(1-v)$, $|J|^{-1} = u$, and

$$f_{U,V}(u,v) = \frac{\lambda e^{-\lambda u}(\lambda u)^{\alpha+\beta-1}}{\Gamma(\alpha+\beta)}\cdot\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}v^{\alpha-1}(1-v)^{\beta-1}.$$

So $U \sim \mathrm{Gamma}(\alpha+\beta,\lambda)$, $V \sim \mathrm{Beta}(\alpha,\beta)$, and they are independent; as a by-product $B(\alpha,\beta) = \Gamma(\alpha)\Gamma(\beta)/\Gamma(\alpha+\beta)$. For instance, if two jobs with exponential durations are done in sequence, the fraction of time spent on the first is uniform and independent of the total.

Example 6.16 (Spacings of exponentials)

For i.i.d. $\mathrm{Exp}(\lambda)$ variables $X_1,\dots,X_n$, the spacings $Y_1 = X_{(1)}$, $Y_j = X_{(j)} - X_{(j-1)}$ are independent with $Y_j \sim \mathrm{Exp}((n-j+1)\lambda)$. Indeed the map from ordered values to spacings has Jacobian 1, and $\sum_j x_{(j)} = \sum_j (n-j+1)y_j$, so the joint density $n!\lambda^n e^{-\lambda\sum x_{(j)}}$ factors. Intuitively, after the first failure the $n-1$ survivors start afresh by memorylessness. Hence $E[X_{(n)}] = \frac{1}{\lambda}\sum_{k=1}^{n}\frac{1}{k}$.

6.7 Exchangeable random variables

$X_1, \dots, X_n$ are **exchangeable** if $(X_{i_1}, \dots, X_{i_n})$ has the same joint law for every permutation. I.i.d. variables are exchangeable, but so are dependent ones such as the indicators X_i that the i-th ball drawn without replacement is white: any particular sequence with k whites has probability $\frac{m(m-1)\cdots(m-k+1)\,(N-m)\cdots(N-m-n+k+1)}{N(N-1)\cdots(N-n+1)}$, which depends only on k. So every draw is white with probability m/N, as in Example 2.1(b). Pólya's urn (draw a ball, return it with another of the same colour) also gives exchangeable draws.

6.8 Selected exercises

Motivation. These are Ross's theoretical exercises for Chapter 6 (book numbering). They practise the three basic operations on joint laws: integrating out to get the law of a function, factorizing a survival function, and conditioning by Bayes's rule in its density form.

Exercise 6.5 (Ratio and product of independent variables)

Let X and Y be independent positive continuous random variables with densities f_X and f_Y.
(a) Show that $Z = X/Y$ has density $f_Z(z) = \int_0^\infty y\, f_X(zy) f_Y(y)\, dy$, $z > 0$.
(b) Show that $W = XY$ has density $f_W(w) = \int_0^\infty \frac{1}{y} f_X(w/y) f_Y(y)\, dy$, $w > 0$.
(c) Evaluate f_Z when $X \sim \mathrm{Exp}(\lambda)$ and $Y \sim \mathrm{Exp}(\mu)$.

Solution.

(a)

Step 1. By independence the joint density is $f_X(x) f_Y(y)$ (Proposition 6.1). For $z > 0$,

$$F_Z(z) = P(X \le zY) = \int_0^\infty \Big(\int_0^{zy} f_X(x)\, dx \Big) f_Y(y)\, dy.$$

Step 2. In the inner integral substitute $x = uy$ ($y > 0$ fixed), so $dx = y\, du$ and u runs over $(0, z)$. Since the integrand is nonnegative, the order of integration may be exchanged (Tonelli):

$$F_Z(z) = \int_0^z \Big(\int_0^\infty y\, f_X(uy) f_Y(y)\, dy \Big)\, du.$$

Hence F_Z is the integral of the bracket, which is therefore a density of Z.

(b) In the same way, $F_W(w) = \int_0^\infty \int_0^{w/y} f_X(x)\, dx\, f_Y(y)\, dy$; substitute $x = u/y$, $dx = du/y$, and exchange the integrals to get $F_W(w) = \int_0^w \int_0^\infty \frac{1}{y} f_X(u/y) f_Y(y)\, dy\, du$.

(c) By (a), for $z > 0$,

$$f_Z(z) = \int_0^\infty y\, \lambda e^{-\lambda z y} \mu e^{-\mu y}\, dy = \frac{\lambda\mu}{(\lambda z + \mu)^2},$$

using $\int_0^\infty y e^{-ay}\, dy = 1/a^2$ with $a = \lambda z + \mu$. ∎

Exercise 6.8 (Hazard rate of a minimum)

Let X and Y be independent positive continuous random variables with hazard rates λ_X and λ_Y, and let $W = \min(X, Y)$.
(a) Express the distribution function of W through F_X and F_Y.
(b) Show that $\lambda_W(t) = \lambda_X(t) + \lambda_Y(t)$.

Solution.

(a) $W > t$ iff $X > t$ and $Y > t$, so by independence $\bar{F}_W(t) = \bar{F}_X(t)\bar{F}_Y(t)$, i.e. $F_W(t) = 1 - \{1 - F_X(t)\}\{1 - F_Y(t)\}$.

(b)

Step 1. At every t where f_X and f_Y are continuous, $\bar{F}_X' = -f_X$ and $\bar{F}_Y' = -f_Y$, so by the product rule

$$f_W(t) = -\bar{F}_W'(t) = f_X(t)\bar{F}_Y(t) + \bar{F}_X(t)f_Y(t).$$

Step 2. Dividing by $\bar{F}_W(t) = \bar{F}_X(t)\bar{F}_Y(t) > 0$ gives $\lambda_W(t) = f_X(t)/\bar{F}_X(t) + f_Y(t)/\bar{F}_Y(t) = \lambda_X(t) + \lambda_Y(t)$.

In particular the minimum of independent $\mathrm{Exp}(\lambda)$ and $\mathrm{Exp}(\mu)$ has constant hazard $\lambda + \mu$, so it is $\mathrm{Exp}(\lambda + \mu)$: competing risks add. ■

Exercise 6.14 (Two geometric waiting times given their sum)

Let X and Y be independent geometric random variables with the same parameter p. Find $P(X = i \mid X + Y = n)$.

Solution.

Step 1. (Guess.) Flip a p-coin repeatedly; X is the time of the first head and $X + Y$ that of the second. Given that the second head is on flip n, the first head should be equally likely to be on any of flips $1, \dots, n-1$.

Step 2. (Verification.) With $q = 1 - p$, for $n \geq 2$ and $1 \leq i \leq n - 1$, independence gives

$$P(X = i,\, Y = n - i) = q^{i-1}p \cdot q^{n-i-1}p = p^2 q^{n-2},$$

which does not depend on i. Summing over the $n - 1$ values of i, $P(X + Y = n) = (n-1)p^2q^{n-2}$, so

$$P(X = i \mid X + Y = n) = \frac{1}{n-1}, \qquad i = 1, \dots, n-1.$$

■

Exercise 6.21 (Poisson counts with a gamma rate)

The moisture W on a day is $\mathrm{Gamma}(t, \lambda)$, and given $W = w$ the number N of accidents that day is $\mathrm{Poisson}(w)$. Show that given $N = n$, W is $\mathrm{Gamma}(t + n, \lambda + 1)$.

Solution.

Step 1. For the mixed pair (W, N) the conditional density of W given $N = n$ is, by Bayes's formula in density form (as in Example 6.12),

$$f_{W|N}(w \mid n) = \frac{P(N = n \mid W = w) f_W(w)}{P(N = n)}, \qquad P(N = n) = \int_0^\infty P(N = n \mid W = w) f_W(w)\, dw.$$

Step 2. For $w > 0$,

$$P(N = n \mid W = w) f_W(w) = e^{-w}\frac{w^n}{n!} \cdot \frac{\lambda e^{-\lambda w}(\lambda w)^{t-1}}{\Gamma(t)} = C_n\, e^{-(\lambda+1)w} w^{t+n-1},$$

where $C_n = \lambda^t / \{n!\,\Gamma(t)\}$ does not depend on w.

Step 3. So $f_{W|N}(w \mid n)$ is proportional to $e^{-(\lambda+1)w}w^{t+n-1}$, which is the kernel of the $\mathrm{Gamma}(t + n, \lambda + 1)$ density. Two densities that are proportional are equal, since both integrate to 1. Hence $W \mid N = n \sim \mathrm{Gamma}(t + n, \lambda + 1)$; in particular $E[W \mid N = n] = (t + n)/(\lambda + 1)$. ■

Lecture 7. Properties of Expectation

7.1 Expectation of sums

For jointly distributed X, Y, $E[g(X,Y)] = \sum_x \sum_y g(x,y)p(x,y)$ or $\iint g(x,y)f(x,y)\,dx\,dy$ (proved as Proposition 5.2). Taking $g = x + y$ gives linearity, $E[X+Y] = E[X] + E[Y]$, and by induction $E[\sum X_i] = \sum E[X_i]$, with no independence needed. Also $X \geq Y$ implies $E[X] \geq E[Y]$.

Example 7.1

(a) An accident at $X \sim U(0,L)$ and an ambulance at independent $Y \sim U(0,L)$: $E|X-Y| = \frac{1}{L^2}\int_0^L \int_0^L |x-y|\,dx\,dy = \frac{L}{3}$.
(b) Boole's inequality: if X counts how many of $A_1, \ldots, A_n$ occur and $Y = \mathbb{1}\{X \geq 1\}$, then $X \geq Y$, so $P(\bigcup A_i) = E[Y] \leq E[X] = \sum P(A_i)$.
(c) The sample mean $\bar{X} = \frac{1}{n}\sum X_i$ of identically distributed variables has $E[\bar{X}] = \mu$.

Example 7.2 (Indicator calculations)

(a) Matches: $E[\#\text{matches}] = N \cdot \frac{1}{N} = 1$ for every N.
(b) Coupon collecting with N types: once i types are held, the number of further purchases until a new type is geometric with success probability $(N-i)/N$. So the total T has

$$E[T] = \sum_{i=0}^{N-1} \frac{N}{N-i} = N\Big(1 + \frac{1}{2} + \cdots + \frac{1}{N}\Big) \approx N \log N.$$

The expected number of distinct types in n purchases is $N\{1-(1-1/N)^n\}$, by summing indicators of "type j appears".
(c) Runs: in a random arrangement of n ones and m zeros, the expected number of runs of ones is $\frac{n(m+1)}{n+m}$. Let I_i be the indicator that a run of ones starts at position i; $P(I_1 = 1) = \frac{n}{n+m}$, and for $i > 1$, $P(I_i = 1) = P(\text{position } i-1 \text{ is } 0,\ i \text{ is } 1) = \frac{mn}{(n+m)(n+m-1)}$.
(d) Quicksort sorts n distinct values by choosing a random pivot, comparing all others with it, and recursing on both sides. The i-th and j-th smallest values ($i < j$) are compared iff one of them is the first pivot chosen among the $j-i+1$ values between them inclusive, which has probability $2/(j-i+1)$. Hence

$$E[\#\text{comparisons}] = \sum_{i<j} \frac{2}{j-i+1} = 2(n+1)\sum_{k=1}^{n} \frac{1}{k} - 4n \approx 2n \log n.$$

Proposition 7.1 (Moments of the number of events)

If X is the number of the events $A_1, \ldots, A_n$ that occur, then

$$E\Big[\binom{X}{k}\Big] = \sum_{i_1 < \cdots < i_k} P(A_{i_1} \cdots A_{i_k}).$$

Proof. $\binom{X}{k}$ is the number of k-subsets of the events that all occur, i.e. $\sum_{i_1<\cdots<i_k} I_{i_1} \cdots I_{i_k}$; take expectations. ∎

With $k = 2$, $E[X^2] = E[X] + 2\sum_{i<j} P(A_i A_j)$. For matches, $P(A_i A_j) = \frac{1}{N(N-1)}$, so $E[X(X-1)] = 1$ and $\mathrm{Var}(X) = 1$, as for Poisson(1).

7.2 Covariance and variance of sums

Motivation. The variance of a sum depends on how the terms move together. Covariance captures this, and expanding $\mathrm{Var}(\sum X_i)$ in covariances solves many problems that look hard, such as matching and sampling without replacement.

Proposition 7.2

If X, Y are independent, $E[g(X)h(Y)] = E[g(X)]E[h(Y)]$.

Proof. $\iint g(x)h(y)f_X(x)f_Y(y)\,dx\,dy$ factors. ∎

$\mathrm{Cov}(X,Y) = E[(X-\mu_X)(Y-\mu_Y)] = E[XY] - E[X]E[Y]$. Independence implies $\mathrm{Cov} = 0$, but not conversely: for X uniform on $\{-1,0,1\}$ and $Y = X^2$, $\mathrm{Cov}(X,Y) = E[X^3] = 0$.

Proposition 7.3

Covariance is symmetric and bilinear: $\mathrm{Cov}(\sum_i a_i X_i, \sum_j b_j Y_j) = \sum_{i,j} a_i b_j \mathrm{Cov}(X_i, Y_j)$, and $\mathrm{Cov}(X,X) = \mathrm{Var}(X)$. Hence

$$\mathrm{Var}\Big(\sum_i X_i\Big) = \sum_i \mathrm{Var}(X_i) + 2\sum_{i<j} \mathrm{Cov}(X_i, X_j),$$

which is $\sum \mathrm{Var}(X_i)$ for pairwise independent variables.

Example 7.3

(a) For i.i.d. X_i with variance σ^2, $\mathrm{Var}(\bar{X}) = \sigma^2/n$, and since $\sum(X_i - \bar{X})^2 = \sum(X_i-\mu)^2 - n(\bar{X}-\mu)^2$,

$$E[S^2] = \frac{1}{n-1}(n\sigma^2 - n\cdot\sigma^2/n) = \sigma^2, \qquad S^2 = \frac{1}{n-1}\sum(X_i - \bar{X})^2.$$

(b) Hypergeometric variance: with X_i the indicator that the i-th draw is white, $\mathrm{Cov}(X_i, X_j) = \frac{m(m-1)}{N(N-1)} - \frac{m^2}{N^2} = -\frac{m(N-m)}{N^2(N-1)}$ for $i \neq j$, which with n draws gives $\mathrm{Var}(X) = n\frac{m}{N}(1-\frac{m}{N})\frac{N-n}{N-1}$.
(c) Multinomial: if N_i is the number of trials with outcome i, then $\mathrm{Cov}(N_i, N_j) = -np_ip_j$ for $i \neq j$: write $N_i = \sum_k I_i(k)$; $I_i(k)I_j(k) = 0$ and trials are independent, so $\mathrm{Cov}(N_i,N_j) = \sum_k \mathrm{Cov}(I_i(k), I_j(k)) = n(0 - p_ip_j)$.

The **correlation** is $\rho(X,Y) = \mathrm{Cov}(X,Y)/\sqrt{\mathrm{Var}(X)\,\mathrm{Var}(Y)}$.

Proposition 7.4

$-1 \le \rho(X,Y) \le 1$, with $\rho = \pm 1$ iff $Y = a + bX$ with probability 1 for some b of the same sign.

Proof. With σ_X^2, σ_Y^2 the variances, $0 \le \mathrm{Var}(X/\sigma_X \pm Y/\sigma_Y) = 2(1 \pm \rho)$. Equality means $X/\sigma_X \pm Y/\sigma_Y$ is constant. ∎

ρ measures *linear* dependence only. For the bivariate normal of Example 6.11, $\rho(X,Y) = \rho$.

7.3 Conditional expectation

Motivation. $E[X \mid Y]$ is the best guess of X once Y is known, itself a random variable. Averaging it gives back $E[X]$, a "divide and conquer" tool for computing expectations by conditioning on a first step.

$E[X \mid Y = y] = \sum_x x p_{X|Y}(x \mid y)$ or $\int x f_{X|Y}(x \mid y)\,dx$. $E[X \mid Y]$ denotes the random variable $g(Y)$ where $g(y) = E[X \mid Y = y]$.

Theorem 7.5 (Tower property)

$E[X] = E[E[X \mid Y]]$, i.e. $E[X] = \sum_y E[X \mid Y = y]P(Y = y)$ or $\int E[X \mid Y = y] f_Y(y)\,dy$.

Proof. Discrete case: $\sum_y \sum_x x p_{X|Y}(x \mid y) p_Y(y) = \sum_x \sum_y x p(x, y) = E[X]$. ■

Example 7.4 (The trapped miner)

A miner chooses at random among three doors: door 1 leads to safety in 3 hours, doors 2 and 3 lead back to the mine after 5 and 7 hours. With X the time to safety and Y the first door, $E[X] = \frac{1}{3}\{3 + (5 + E[X]) + (7 + E[X])\}$, so $E[X] = 15$.

Example 7.5 (Random sums)

If N is a nonnegative integer variable independent of i.i.d. $X_1, X_2, \ldots$, then $E[\sum_{i=1}^N X_i \mid N = n] = nE[X]$, so

$$E\Big[\sum_{i=1}^{N} X_i\Big] = E[N]E[X].$$

(The expected spending in a store is the expected number of customers times the mean spending.)

Example 7.6 (Waiting for k consecutive successes)

In independent trials with success probability p, let N_k be the number of trials until k consecutive successes. Conditioning on N_{k-1}: one more trial either succeeds (done) or fails (start over), so $E[N_k \mid N_{k-1}] = N_{k-1} + 1 + (1 - p)E[N_k]$, giving $E[N_k] = \frac{1}{p} + \frac{E[N_{k-1}]}{p}$ and

$$E[N_k] = \frac{1}{p} + \frac{1}{p^2} + \cdots + \frac{1}{p^k}.$$

For a fair coin, $E[N_2] = 6$ and $E[N_3] = 14$.

Probabilities by conditioning. With $X = I_E$, the tower property reads $P(E) = E[P(E \mid Y)]$. For instance, for independent continuous X, Y, $P(X < Y) = \int F_X(y) f_Y(y)\,dy$, and $P(X + Y < a) = \int F_X(a - y) f_Y(y)\,dy$ is the convolution formula again.

Example 7.7 (The best-prize problem)

n prizes of distinct values appear in random order; each must be accepted or rejected on the spot, knowing only the relative ranks so far. Strategy: reject the first k, then accept the first prize better than all earlier ones. Let X be the position of the best prize. If $X = i > k$, we

win iff the best of the first $i-1$ prizes is among the first k, which has probability $k/(i-1)$. So

$$P_k(\text{best}) = \frac{1}{n}\sum_{i=k+1}^{n}\frac{k}{i-1} \approx \frac{k}{n}\log\frac{n}{k}.$$

Maximizing $x\log(1/x)$ over $x = k/n$ gives $k \approx n/e$, and the success probability tends to $1/e \approx 0.368$ however large n is.

Example 7.8 (Uniform sums revisited)

Let $m(x) = E[N(x)]$, where $N(x) = \min\{n : U_1 + \cdots + U_n > x\}$ for $0 \le x \le 1$. Conditioning on $U_1 = y$: if $y > x$ then $N = 1$, otherwise $N = 1 + N'(x-y)$. So $m(x) = 1 + \int_0^x m(x-y)\,dy = 1 + \int_0^x m(u)\,du$, whence $m' = m$, $m(0) = 1$, and $m(x) = e^x$. At $x = 1$ this recovers Example 6.8.

Proposition 7.6 (Conditional variance formula)

With $\operatorname{Var}(X \mid Y) = E[(X - E[X \mid Y])^2 \mid Y]$,

$$\operatorname{Var}(X) = E\big[\operatorname{Var}(X \mid Y)\big] + \operatorname{Var}\big(E[X \mid Y]\big).$$

Proof. $E[\operatorname{Var}(X \mid Y)] = E[X^2] - E[(E[X \mid Y])^2]$ and $\operatorname{Var}(E[X \mid Y]) = E[(E[X \mid Y])^2] - (E[X])^2$; add. ∎

Example 7.9

(a) For the random sum $S = \sum_1^N X_i$: $E[S \mid N] = N\mu$ and $\operatorname{Var}(S \mid N) = N\sigma^2$, so $\operatorname{Var}(S) = E[N]\sigma^2 + \mu^2\operatorname{Var}(N)$. With $N \sim \text{Poisson}(\lambda)$, $\operatorname{Var}(S) = \lambda E[X^2]$.
(b) Geometric variance: if Y indicates success on the first trial, $E[N \mid Y]$ is 1 or $1 + 1/p$ and $\operatorname{Var}(N \mid Y)$ is 0 or $\operatorname{Var}(N)$; solving gives $\operatorname{Var}(N) = (1-p)/p^2$.

7.4 Conditional expectation and prediction

Motivation. Among all functions of Y, $E[X \mid Y]$ minimizes mean squared prediction error. When we restrict to linear functions, we obtain the regression line.

Proposition 7.7

$E[(Y - g(X))^2] \ge E[(Y - E[Y \mid X])^2]$ for every function g: $E[Y \mid X]$ is the best predictor of Y from X in mean square error.

Proof. Given X, $E[(Y-c)^2 \mid X]$ is minimized by $c = E[Y \mid X]$ (variance $\le$ any second moment about c); take expectations. ∎

If we restrict to linear predictors $a + bX$, minimizing $E[(Y - a - bX)^2]$ gives

$$b = \rho\frac{\sigma_Y}{\sigma_X}, \qquad a = \mu_Y - b\mu_X, \qquad \text{MSE} = \sigma_Y^2(1-\rho^2).$$

For the bivariate normal $E[Y \mid X]$ is linear, so the best linear predictor is the best predictor. Galton's observation that sons of tall fathers are on average less tall ("regression to the mean") is the factor $|\rho| < 1$.

7.5 Moment generating functions

Motivation. The mgf turns sums of independent variables into products and determines the distribution. This makes it the main tool for identifying the distribution of a sum and for proving the central limit theorem.

The **moment generating function** of X is $M(t) = E[e^{tX}]$. When finite near 0, $M^{(n)}(0) = E[X^n]$ (differentiate under the expectation).

Distribution	$M(t)$	valid for
Bin(n,p)	$(pe^t + 1-p)^n$	all t
Poisson(λ)	$\exp\{\lambda(e^t-1)\}$	all t
Geometric(p)	$pe^t/\{1-(1-p)e^t\}$	$(1-p)e^t < 1$
Exp(λ)	$\lambda/(\lambda - t)$	$t < \lambda$
Gamma(α,λ)	$\{\lambda/(\lambda-t)\}^\alpha$	$t < \lambda$
$N(\mu,\sigma^2)$	$\exp\{\mu t + \sigma^2 t^2/2\}$	all t

For the normal: $E[e^{tZ}] = \frac{1}{\sqrt{2\pi}} \int e^{-(x-t)^2/2 + t^2/2}\,dx = e^{t^2/2}$, and $X = \mu + \sigma Z$.

Theorem 7.8

(a) If X, Y are independent, $M_{X+Y}(t) = M_X(t) M_Y(t)$.
(b) (Uniqueness) If $M_X = M_Y$ on a neighbourhood of 0, then X and Y have the same distribution.

(a) is Proposition 7.2 with $g = e^{tx}$, $h = e^{ty}$;

(b) is quoted without proof. Together they make Proposition 6.2 immediate: e.g. $e^{\mu_1 t + \sigma_1^2 t^2/2} e^{\mu_2 t + \sigma_2^2 t^2/2}$ is the normal mgf with parameters $\mu_1 + \mu_2$, $\sigma_1^2 + \sigma_2^2$.

Example 7.10

(a) χ^2_n: $E[e^{tZ^2}] = (1-2t)^{-1/2}$, so $Z_1^2 + \cdots + Z_n^2$ has mgf $(1-2t)^{-n/2}$, the Gamma$(n/2, 1/2)$ mgf.
(b) Random sums: $E[e^{tS} \mid N] = M_X(t)^N$, so $M_S(t) = E[M_X(t)^N]$. Differentiating at 0 recovers $E[S] = E[N]\mu$ and the variance formula of Example 7.9.
(c) If $M(t) = e^{3(e^t-1)}$, then $X \sim$ Poisson(3) and $P(X=0) = e^{-3}$.

The **joint mgf** $M(t_1,\ldots,t_n) = E[e^{t_1X_1 + \cdots + t_nX_n}]$ determines the joint law, and $X_1,\ldots,X_n$ are independent iff it factors into the product of the marginal mgfs.

Example 7.11

Let X, Y be i.i.d. $N(\mu,\sigma^2)$. Then $X+Y$ and $X-Y$ are independent: $t(X+Y) + s(X-Y) = (t+s)X + (t-s)Y$, so

$$E[e^{t(X+Y)+s(X-Y)}] = e^{2\mu t + \sigma^2(t^2+s^2)} = E[e^{t(X+Y)}]\,E[e^{s(X-Y)}].$$

7.6 The multivariate normal distribution

If $Z_1, \dots, Z_n$ are independent $N(0,1)$ and $X_i = \mu_i + \sum_j a_{ij} Z_j$ $(i = 1, \dots, m)$, then $(X_1, \dots, X_m)$ is **multivariate normal**. Every linear combination $\sum t_i X_i$ is then normal, with mean $\sum t_i \mu_i$ and variance $\sum_{i,j} t_i t_j \operatorname{Cov}(X_i, X_j)$, so

$$M(t_1, \dots, t_m) = \exp\Big\{ \sum_i t_i \mu_i + \frac{1}{2} \sum_{i,j} t_i t_j \operatorname{Cov}(X_i, X_j) \Big\} :$$

the joint law is determined by the means and covariances. In particular, uncorrelated components of a multivariate normal vector are independent (the mgf factors).

Theorem 7.9 (Sample mean and variance)

If $X_1, \dots, X_n$ are i.i.d. $N(\mu, \sigma^2)$, then $\bar{X}$ and S^2 are independent, $\bar{X} \sim N(\mu, \sigma^2/n)$, and $(n-1)S^2/\sigma^2 \sim \chi^2_{n-1}$.

Proof. $(\bar{X}, X_1 - \bar{X}, \dots, X_n - \bar{X})$ is multivariate normal, and $\operatorname{Cov}(\bar{X}, X_i - \bar{X}) = \sigma^2/n - \sigma^2/n = 0$, so $\bar{X}$ is independent of the deviations and hence of S^2. Next,

$$\sum_{i=1}^{n} \Big(\frac{X_i - \mu}{\sigma} \Big)^2 = \frac{(n-1)S^2}{\sigma^2} + \Big(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \Big)^2 .$$

The left side is χ^2_n and the last term is χ^2_1 independent of the first; dividing mgfs, $(n-1)S^2/\sigma^2$ has mgf $(1-2t)^{-n/2}/(1-2t)^{-1/2} = (1-2t)^{-(n-1)/2}$. ∎

7.7 General definition of expectation

For a nonnegative X that is neither discrete nor continuous one defines $E[X]$ as the limit of $E[X_n]$ for simple $X_n \uparrow X$ (the Lebesgue integral), or as $\int_0^\infty P(X > x)\, dx$; for general X, $E[X] = E[X^+] - E[X^-]$. All properties above continue to hold. If $X = Y$ with probability p and W otherwise, $E[X] = pE[Y] + (1-p)E[W]$.

7.8 Selected exercises

Motivation. These are Ross's theoretical exercises for Chapter 7 (book numbering). They exercise indicator variables, conditioning on the right quantity, and the conditional variance formula, ending with the mean and variance of a branching process.

Exercise 7.13 (Record values)

Let $X_1, \dots, X_n$ be i.i.d. continuous random variables. A record occurs at time j if $X_j \geq X_i$ for all $i \leq j$. Show that
(a) $E[\text{number of records}] = \sum_{j=1}^{n} 1/j$;
(b) $\operatorname{Var}(\text{number of records}) = \sum_{j=1}^{n} (j-1)/j^2$.

Solution.

Step 1. (Ranks are uniform.) For $i \neq k$, $P(X_i = X_k) = 0$ since X_i is continuous and independent of X_k; so with probability 1 the values are distinct. For $m \leq n$, the joint density of $(X_1, \dots, X_m)$ is symmetric, so each of the $m!$ orderings of $X_1, \dots, X_m$ has the same probability $1/m!$.

Step 2. Let I_j be the indicator of a record at time j, i.e. of $X_j = \max(X_1, \dots, X_j)$. Of the $j!$ orderings of $X_1, \dots, X_j$, exactly $(j-1)!$ put X_j on top, so $P(I_j = 1) = 1/j$. Part (a) follows from $E[\sum_j I_j] = \sum_j 1/j$.

Step 3. Let $i < j$. Among the $j!$ equally likely orderings of $X_1, \dots, X_j$, those with X_j on top correspond to the $(j-1)!$ orderings of $X_1, \dots, X_{j-1}$; of these, a fraction $1/i$ puts X_i above $X_1, \dots, X_{i-1}$ (apply Step 1 to the first i variables). So $P(I_i I_j = 1) = \frac{(j-1)!/i}{j!} = \frac{1}{ij} = P(I_i = 1)P(I_j = 1)$, and $\operatorname{Cov}(I_i, I_j) = 0$.

Step 4. By Proposition 7.3 and Step 3,

$$\operatorname{Var}\Big(\sum_{j=1}^{n} I_j\Big) = \sum_{j=1}^{n} \operatorname{Var}(I_j) = \sum_{j=1}^{n} \frac{1}{j}\Big(1 - \frac{1}{j}\Big) = \sum_{j=1}^{n} \frac{j-1}{j^2}.$$

■

Exercise 7.20 (The conditional covariance formula)

For X, Y, Z with $E[X^2], E[Y^2] < \infty$ define $\operatorname{Cov}(X, Y \mid Z) = E\big[(X - E[X \mid Z])(Y - E[Y \mid Z]) \mid Z\big]$.
(a) Show that $\operatorname{Cov}(X, Y \mid Z) = E[XY \mid Z] - E[X \mid Z]E[Y \mid Z]$.
(b) Prove that $\operatorname{Cov}(X, Y) = E[\operatorname{Cov}(X, Y \mid Z)] + \operatorname{Cov}(E[X \mid Z], E[Y \mid Z])$.
(c) Deduce the conditional variance formula.

Solution.

(a) Given $Z = z$, a function $g(Z)$ equals the constant $g(z)$, so $E[g(Z)V \mid Z] = g(Z)E[V \mid Z]$ for any integrable V. Expanding the product and using this with $g(Z) = E[X \mid Z]$ and $g(Z) = E[Y \mid Z]$,

$$\begin{aligned}\operatorname{Cov}(X, Y \mid Z) &= E[XY \mid Z] - E[X \mid Z]E[Y \mid Z] - E[Y \mid Z]E[X \mid Z] + E[X \mid Z]E[Y \mid Z] \\ &= E[XY \mid Z] - E[X \mid Z]E[Y \mid Z].\end{aligned}$$

(b) Take expectations in (a) and use the tower property (Theorem 7.5):

$$E[\operatorname{Cov}(X, Y \mid Z)] = E[XY] - E\big[E[X \mid Z]E[Y \mid Z]\big].$$

Since $E\big[E[X \mid Z]\big] = E[X]$ and likewise for Y,

$$\operatorname{Cov}(E[X \mid Z], E[Y \mid Z]) = E\big[E[X \mid Z]E[Y \mid Z]\big] - E[X]E[Y].$$

Adding the two displays gives $E[XY] - E[X]E[Y] = \operatorname{Cov}(X, Y)$.

(c) With $Y = X$, $\operatorname{Cov}(X, X \mid Z) = \operatorname{Var}(X \mid Z)$ and (b) becomes $\operatorname{Var}(X) = E[\operatorname{Var}(X \mid Z)] + \operatorname{Var}(E[X \mid Z])$, which is Proposition 7.6. ■

Exercise 7.35 (Generating functions as probabilities)

Let X be nonnegative integer valued with $p_j = P(X = j)$ and generating function $\phi(s) = E[s^X] = \sum_j p_j s^j$. For $0 < s < 1$ let Y be geometric with parameter $1 - s$, independent of X. Show that $\phi(s) = P(X < Y)$.

Solution.

Step 1. $P(Y = k) = s^{k-1}(1-s)$ for $k \geq 1$, so $P(Y > j) = s^j$ for $j \geq 0$ (the first j trials all fail).

Step 2. Conditioning on X and using independence,

$$P(X < Y) = \sum_{j=0}^{\infty} P(X = j)P(Y > j) = \sum_{j=0}^{\infty} p_j s^j = \phi(s).$$

So $\phi(s)$ is the probability that X is smaller than an independent geometric clock whose trials fail with probability s. ■

Exercise 7.44 (Moments of a branching process)

Each individual of a population produces j offspring with probability P_j, independently of all others; $\mu = \sum_j jP_j$ and $\sigma^2 = \sum_j (j-\mu)^2 P_j < \infty$. Let X_n be the size of generation n, with $X_0 = 1$. Show that
(a) $E[X_n] = \mu E[X_{n-1}]$, hence $E[X_n] = \mu^n$;
(b) $\operatorname{Var}(X_n) = \sigma^2 \mu^{n-1} + \mu^2 \operatorname{Var}(X_{n-1})$;
(c) $\operatorname{Var}(X_n) = \sigma^2 \mu^{n-1} \dfrac{\mu^n - 1}{\mu - 1}$ if $\mu \neq 1$, and $\operatorname{Var}(X_n) = n\sigma^2$ if $\mu = 1$.

Solution.

Step 1. Write $X_n = \sum_{i=1}^{X_{n-1}} Z_i$, where Z_i is the number of offspring of the i-th member of generation $n-1$. The Z_i are i.i.d. with mean μ and variance σ^2, and independent of X_{n-1}. Given $X_{n-1} = k$, X_n is a sum of k independent such variables, so

$$E[X_n \mid X_{n-1}] = \mu X_{n-1}, \qquad \operatorname{Var}(X_n \mid X_{n-1}) = \sigma^2 X_{n-1}.$$

(a) By the tower property, $E[X_n] = \mu E[X_{n-1}]$, and since $E[X_0] = 1$, induction gives $E[X_n] = \mu^n$.

(b) By the conditional variance formula (Proposition 7.6) and (a),

$$\operatorname{Var}(X_n) = E[\sigma^2 X_{n-1}] + \operatorname{Var}(\mu X_{n-1}) = \sigma^2 \mu^{n-1} + \mu^2 \operatorname{Var}(X_{n-1}).$$

(c) Let $V_n = \operatorname{Var}(X_n)$; $V_0 = 0$ since $X_0 = 1$. If $\mu = 1$, (b) reads $V_n = \sigma^2 + V_{n-1}$, so $V_n = n\sigma^2$. If $\mu \neq 1$, the formula holds for $n = 0$, and if it holds for $n-1$ then

$$\begin{aligned} V_n &= \sigma^2 \mu^{n-1} + \mu^2 \sigma^2 \mu^{n-2} \frac{\mu^{n-1} - 1}{\mu - 1} = \sigma^2 \mu^{n-1} \frac{(\mu - 1) + \mu(\mu^{n-1} - 1)}{\mu - 1} \\ &= \sigma^2 \mu^{n-1} \frac{\mu^n - 1}{\mu - 1}. \end{aligned}$$

■

Lecture 8. Limit Theorems

8.1 Markov and Chebyshev inequalities; the weak law

Motivation. We often know only a mean or variance. Markov's and Chebyshev's inequalities give bounds on tail probabilities from these alone, and Chebyshev immediately shows that sample averages concentrate around the mean.

Proposition 8.1 (Markov's inequality)

If $X \geq 0$ and $a > 0$, then $P(X \geq a) \leq E[X]/a$.

Proof. $a\mathbb{1}\{X \geq a\} \leq X$; take expectations. ■

Proposition 8.2 (Chebyshev's inequality)

If X has mean μ and variance σ^2, then $P(|X - \mu| \geq k) \leq \sigma^2/k^2$ for $k > 0$.

Proof. Apply Markov's inequality to $(X - \mu)^2$ and $a = k^2$. ■

These bounds use only the mean (and variance), so they are often crude, but they hold for every distribution.

Example 8.1

(a) Weekly output has mean 50. Then $P(X \geq 75) \leq 50/75 = 2/3$. If also $\mathrm{Var}(X) = 25$, $P(40 < X < 60) \geq 1 - 25/100 = 3/4$.
(b) For $X \sim U(0, 10)$, Chebyshev gives $P(|X - 5| > 4) \leq (25/3)/16 \approx 0.52$, whereas the true value is 0.2.
(c) If $\mathrm{Var}(X) = 0$, then $P(|X - \mu| > 1/n) = 0$ for all n, and by continuity $P(X = \mu) = 1$.

Theorem 8.3 (Weak law of large numbers)

If $X_1, X_2, \ldots$ are i.i.d. with mean μ, then for every $\epsilon > 0$, $P(|\bar{X}_n - \mu| \geq \epsilon) \to 0$ as $n \to \infty$.

Proof. Assuming finite variance σ^2: $E[\bar{X}_n] = \mu$, $\mathrm{Var}(\bar{X}_n) = \sigma^2/n$, and Chebyshev gives the bound $\sigma^2/(n\epsilon^2) \to 0$. ■

8.2 The central limit theorem

Motivation. The law of large numbers says averages converge; the central limit theorem says how fluctuations around the mean look: approximately normal, whatever the original distribution. This is why the normal distribution is everywhere in statistics.

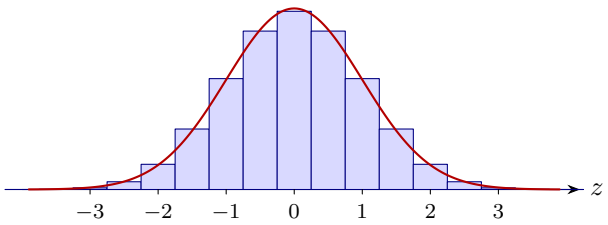


Figure. *The distribution of the standardized sum* $(S_{16}-8)/2$ *for* $S_{16}\sim\mathrm{Bin}(16,\frac12)$ *(bars, scaled to unit area) against the standard normal density. Already for* $n=16$ *the CLT approximation is close.*

Theorem 8.4 (Central limit theorem)

If $X_1, X_2, \dots$ are i.i.d. with mean μ and variance $\sigma^2 \in (0,\infty)$, then for every a,

$$P\Big(\frac{X_1+\cdots+X_n-n\mu}{\sigma\sqrt{n}} \le a\Big) \to \Phi(a) \qquad (n\to\infty).$$

The key tool is: if $M_{Z_n}(t) \to M_Z(t)$ for all t, then $F_{Z_n}(a) \to F_Z(a)$ at continuity points of F_Z (quoted).

Proof sketch. Take $\mu=0$, $\sigma=1$, and assume the mgf M of X_i exists near 0. Let $L(t)=\log M(t)$; then $L(0)=0$, $L'(0)=\mu=0$, $L''(0)=E[X^2]-\mu^2=1$. The mgf of $Z_n=\sum X_i/\sqrt{n}$ is $M(t/\sqrt{n})^n$, and by L'Hôpital twice (in $x=1/\sqrt{n}$),

$$nL\Big(\frac{t}{\sqrt{n}}\Big) = \frac{L(tx)}{x^2} \to \frac{t^2 L''(0)}{2} = \frac{t^2}{2}.$$

So $M_{Z_n}(t)\to e^{t^2/2}$, the $N(0,1)$ mgf. (Without an mgf the same argument works with characteristic functions.) ∎

The theorem explains why measurement errors, heights, and sums of many small effects are approximately normal. It also holds for independent, non-identically distributed X_i that are uniformly bounded with $\sum \mathrm{Var}(X_i)=\infty$: then $(\sum(X_i-\mu_i))/\sqrt{\sum\sigma_i^2}$ is approximately $N(0,1)$.

Example 8.2 (How many measurements?)

An astronomer averages n independent measurements with variance 4. To be 95% sure that the average is within 0.5 of the true value, the CLT requires $P(|Z|\le 0.5\sqrt{n}/2)\approx 0.95$, i.e. $0.25\sqrt{n}\ge 1.96$, $n\ge 62$. Chebyshev's inequality would demand $4/(0.25n)\le 0.05$, i.e. $n\ge 320$: guaranteed for every distribution, but far more conservative.

Example 8.3

(a) The sum S of 10 fair dice has mean 35 and variance $350/12$; with the continuity correction, $P(30\le S\le 40)\approx 2\Phi(5.5/5.40)-1\approx 0.69$.
(b) The sum of 10 independent $U(0,1)$ variables has mean 5 and variance $10/12$, so $P(\sum X_i > 6)\approx 1-\Phi(1.10)\approx 0.137$.
(c) Exam grading times are i.i.d. with mean 20 and standard deviation 4 minutes. The probability that the first 25 exams are graded within 450 minutes is approximately $\Phi((450-500)/20)=\Phi(-2.5)\approx 0.006$.

8.3 The strong law of large numbers

Motivation. The weak law concerns each fixed n; the strong law says that the sequence of averages itself converges with probability one, which justifies the frequency interpretation of probability.

Theorem 8.5 (Strong law of large numbers)

If $X_1, X_2, \ldots$ are i.i.d. with finite mean μ, then $P(\lim_{n\to\infty} \bar{X}_n = \mu) = 1$.

Proof. Under the extra assumption $K = E[X_i^4] < \infty$, with $\mu = 0$. Expanding $E[S_n^4]$, the terms with some index appearing exactly once vanish, leaving $nE[X^4] + 6\binom{n}{2}E[X_i^2 X_j^2] \le nK + 3n(n-1)K$ (as $E[X^2]^2 \le E[X^4]$). Hence $E[\sum_n S_n^4/n^4] \le \sum_n (K/n^3 + 3K/n^2) < \infty$, so $\sum_n S_n^4/n^4 < \infty$ with probability 1, which forces $S_n^4/n^4 \to 0$, i.e. $\bar{X}_n \to 0$. ■

The weak law says that for each large n, $\bar{X}_n$ is *probably* close to μ; the strong law says that, with probability 1, the whole sequence eventually stays close to μ. Applied to $X_i = \mathbb{1}\{E \text{ occurs on trial } i\}$, it justifies the frequency interpretation: the long-run proportion of trials on which E occurs equals $P(E)$ with probability 1.

8.4 Other inequalities

Motivation. Sharper or one-sided bounds are often needed: Chernoff bounds give exponential tails from the mgf, and Jensen's inequality compares $E[g(X)]$ with $g(E[X])$ for convex g.

Proposition 8.6 (One-sided Chebyshev)

If $E[X] = 0$ and $\mathrm{Var}(X) = \sigma^2$, then $P(X \ge a) \le \sigma^2/(\sigma^2 + a^2)$ for $a > 0$.

Proof. For $b > 0$, $P(X \ge a) \le P((X+b)^2 \ge (a+b)^2) \le (\sigma^2 + b^2)/(a+b)^2$ by Markov; choose $b = \sigma^2/a$. ■

Proposition 8.7 (Chernoff bounds)

If $M(t) = E[e^{tX}]$, then $P(X \ge a) \le e^{-ta}M(t)$ for all $t > 0$ and $P(X \le a) \le e^{-ta}M(t)$ for all $t < 0$.

Proof. For $t > 0$, $P(X \ge a) = P(e^{tX} \ge e^{ta}) \le E[e^{tX}]e^{-ta}$ by Markov. ■

One then minimizes over t.

Example 8.4

(a) Standard normal: $e^{-ta+t^2/2}$ is minimized at $t = a$, so $P(Z \ge a) \le e^{-a^2/2}$ for $a > 0$.
(b) Poisson(λ) and $i > \lambda$: minimizing $e^{\lambda(e^t-1)-ti}$ at $e^t = i/\lambda$ gives $P(X \ge i) \le e^{-\lambda}(e\lambda)^i/i^i$.
(c) A gambler wins or loses 1 with probability $\frac{1}{2}$ each play. After n plays, S_n has $M(t) = \cosh(t)^n \le e^{nt^2/2}$, so $P(S_n \ge a) \le e^{-a^2/(2n)}$ (take $t = a/n$); e.g. $P(S_{10} \ge 6) \le e^{-1.8} \approx 0.165$, while the exact value is 0.0547.

Proposition 8.8 (Jensen's inequality)

If f is convex, then $E[f(X)] \geq f(E[X])$ whenever the expectations exist.

Proof. With $\mu = E[X]$, convexity gives $f(x) \geq f(\mu) + f'(\mu)(x-\mu)$ (for twice differentiable f, by Taylor's theorem with $f'' \geq 0$). Take expectations. ∎

For example, an investor choosing between a sure return μ and a random return X with mean μ, with concave utility u, prefers the sure amount since $E[u(X)] \leq u(\mu)$.

8.5 The Poisson approximation, quantified

Let $X = \sum_{i=1}^{n} X_i$ with independent $X_i \sim \text{Bernoulli}(p_i)$, and $Y \sim \text{Poisson}(\lambda)$ with $\lambda = \sum p_i$. Coupling each X_i with a $\text{Poisson}(p_i)$ variable Y_i so that they agree with probability at least $1 - p_i^2$, one obtains, for every set A of integers,

$$|P(X \in A) - P(Y \in A)| \leq \sum_{i=1}^{n} p_i^2.$$

For $\text{Bin}(n,p)$ the bound is $np^2 = \lambda p$: the approximation is good when p is small, whatever n is.

Example 8.5 (Birthdays, again)

Let $X_1, X_2, \ldots$ be the birthdays of people arriving in sequence and N the number needed for the first shared birthday. For n people, the event $N > n$ means that none of the trials "person j shares with one of persons $1, \ldots, j-1$" succeeds; these have success probabilities $(j-1)/365$ given no earlier success, and an exact product gives $P(N > n) = \prod_{j=1}^{n}(1 - \frac{j-1}{365}) \approx \exp\{-\sum_j \frac{j-1}{365}\} = e^{-n(n-1)/730}$, the Poisson-paradigm answer of Example 4.6.

8.6 Selected exercises

Motivation. These are Ross's theoretical exercises for Chapter 8 (book numbering). The first two turn the weak law into a proof of the Weierstrass approximation theorem; the last two give a Markov-type bound for monotone distributions and the optimized Chernoff bound for the binomial.

Exercise 8.4 (Convergence in probability and bounded functions)

Let $Z_1, Z_2, \ldots$ be random variables and c a constant such that $P(|Z_n - c| > \epsilon) \to 0$ for every $\epsilon > 0$. Show that $E[g(Z_n)] \to g(c)$ for every bounded continuous g.

Solution.

Step 1. Let $|g| \leq M$ and fix $\epsilon > 0$. By continuity at c there is $\delta > 0$ with $|g(x) - g(c)| < \epsilon$ whenever $|x - c| \leq \delta$.

Step 2. Split according to whether $|Z_n - c| \leq \delta$. On that event $|g(Z_n) - g(c)| < \epsilon$, and on its complement $|g(Z_n) - g(c)| \leq 2M$. Hence

$$|E[g(Z_n)] - g(c)| \leq E|g(Z_n) - g(c)| \leq \epsilon + 2M\, P(|Z_n - c| > \delta).$$

Step 3. The last probability tends to 0, so $\limsup_n |E[g(Z_n)] - g(c)| \leq \epsilon$. As ϵ is arbitrary, $E[g(Z_n)] \to g(c)$. ∎

Exercise 8.5 (Bernstein polynomials)

Let f be continuous on $[0,1]$ and

$$B_n(x) = \sum_{k=0}^{n} f\Big(\frac{k}{n}\Big)\binom{n}{k} x^k (1-x)^{n-k}.$$

Prove that $B_n(x) \to f(x)$, uniformly in $x \in [0,1]$. (This is a probabilistic proof of the Weierstrass approximation theorem.)

Solution.

Step 1. Fix $x \in [0,1]$ and let $S_n \sim \mathrm{Bin}(n,x)$, the number of successes in n independent trials with success probability x. Then $B_n(x) = E[f(S_n/n)]$, and $E[S_n/n] = x$, $\mathrm{Var}(S_n/n) = x(1-x)/n \le 1/(4n)$.

Step 2. f is bounded, say $|f| \le M$, and uniformly continuous on $[0,1]$: given $\epsilon > 0$ there is $\delta > 0$ with $|f(u) - f(v)| < \epsilon$ whenever $|u - v| \le \delta$.

Step 3. As in Exercise 8.4 (with $c = x$; note $S_n/n \in [0,1]$, where f is defined) and by Chebyshev's inequality (Proposition 8.2),

$$|B_n(x) - f(x)| \le \epsilon + 2M\, P\Big(\Big|\frac{S_n}{n} - x\Big| > \delta\Big) \le \epsilon + \frac{2M}{4n\delta^2}.$$

Step 4. The right side does not depend on x, so $\limsup_n \sup_{x\in[0,1]} |B_n(x) - f(x)| \le \epsilon$ for every $\epsilon > 0$. Hence $B_n \to f$ uniformly on $[0,1]$, and every continuous function on $[0,1]$ is a uniform limit of polynomials. ■

Exercise 8.6(a) (Monotone probability mass functions)

Let X take values in $\{1, 2, \dots\}$ with $P(X = k)$ nonincreasing in k. Prove that $P(X = k) \le 2E[X]/k^2$.

Solution.

Step 1. Let $p_i = P(X = i)$. Dropping the terms with $i > k$, which are nonnegative,

$$E[X] = \sum_{i\ge 1} i p_i \ge \sum_{i=1}^{k} i p_i.$$

Step 2. Since $p_i \ge p_k$ for $i \le k$, $\sum_{i=1}^{k} i p_i \ge p_k \sum_{i=1}^{k} i = p_k \frac{k(k+1)}{2} \ge p_k \frac{k^2}{2}$. Combining, $p_k \le 2E[X]/k^2$. ■

Exercise 8.11 (The Chernoff bound for the binomial)

Let $X \sim \mathrm{Bin}(n,p)$, $0 < p < 1$, $q = 1 - p$, and $np < i < n$.
(a) Show that $\min_{t>0} e^{-ti} E[e^{tX}]$ is attained where $e^t = iq/\{(n-i)p\}$.
(b) Deduce

$$P(X \ge i) \le \frac{n^n}{i^i (n-i)^{n-i}}\, p^i q^{n-i}.$$

Solution.

(a)

Step 1. $E[e^{tX}] = (pe^t + q)^n$, so we minimize $h(t) = -ti + n\log(pe^t + q)$ over $t > 0$. Here

$$h'(t) = -i + \frac{npe^t}{pe^t + q}, \qquad h''(t) = \frac{npqe^t}{(pe^t + q)^2} > 0,$$

so h is strictly convex and any zero of h' is its global minimum.

Step 2. $h'(t) = 0$ iff $npe^t = i(pe^t + q)$, i.e. $e^t = iq/\{(n-i)p\}$ (which needs $i < n$). This t is positive: $iq > (n-i)p$ is equivalent to $i > np$.

(b) At this t, $pe^t + q = \frac{iq}{n-i} + q = \frac{nq}{n-i}$ and $e^{-ti} = \left(\frac{(n-i)p}{iq}\right)^i$. By Proposition 8.7,

$$P(X \geq i) \leq \Big(\frac{(n-i)p}{iq}\Big)^i \Big(\frac{nq}{n-i}\Big)^n = \frac{n^n}{i^i (n-i)^{n-i}}\, p^i q^{n-i}.$$

For instance, with $n = 100$, $p = \frac{1}{2}$ and $i = 60$ the bound is about 0.134; the exact value is about 0.028. ■

Lecture 9. Additional Topics in Probability

9.1 The Poisson process

Motivation. Events occurring randomly in time, with independent increments and no simultaneous events, must form a Poisson process. Counts in intervals are Poisson and the waiting times between events are exponential.

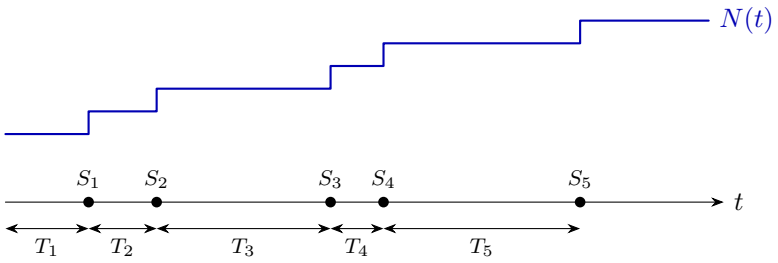


Figure. *A Poisson process: arrival times $S_1 < S_2 < \cdots$, interarrival times $T_k = S_k - S_{k-1}$ i.i.d.* Exp(λ), *and the counting process $N(t)$ (step function, one step per arrival) with $N(t) \sim$* Poisson(λt).

Definition (Poisson process)

A family $\{N(t), t \geq 0\}$ counting events in $[0, t]$ is a **Poisson process with rate** $\lambda > 0$ if (i) $N(0) = 0$;
(ii) the numbers of events in disjoint intervals are independent (independent increments);
(iii) the law of the number of events in $(s, s+t]$ depends only on t (stationary increments);
(iv) $P(N(h) = 1) = \lambda h + o(h)$ and $P(N(h) \geq 2) = o(h)$.

Proposition 9.1

For a Poisson process, $N(t) \sim$ Poisson(λt), and more generally the number of events in any interval of length t is Poisson(λt).

Proof. Split $[0, t]$ into n pieces of length t/n. By (iv) and Boole's inequality, the probability that some piece has two or more events is at most $n\, o(t/n) \to 0$. Ignoring this, $N(t)$ is the number of pieces with an event, which by (ii)–(iv) is Bin($n, \lambda t/n + o(t/n)$), and Proposition 4.4 gives the Poisson(λt) limit. ∎

Let T_1 be the time of the first event and T_n the time between the $(n-1)$-th and n-th events.

Proposition 9.2

The interarrival times $T_1, T_2, \ldots$ are i.i.d. Exp(λ). Hence the time $S_n = T_1 + \cdots + T_n$ of the n-th event is Gamma(n, λ).

Proof. $P(T_1 > t) = P(N(t) = 0) = e^{-\lambda t}$. Given $T_1 = s$, the event $T_2 > t$ means no event in $(s, s+t]$, which by independent and stationary increments has probability $e^{-\lambda t}$ regardless of s; so T_2 is Exp(λ) and independent of T_1. Continue inductively. The gamma law is Proposition 6.2(a), or directly $P(S_n \leq t) = P(N(t) \geq n)$ as in Lecture 5. ∎

Conversely, starting from i.i.d. exponential interarrival times produces a Poisson process; memorylessness is what makes the increments independent.

Example 9.1

Earthquakes occur as a Poisson process with rate 2 per week. The probability of at least 3 in the next 2 weeks is $1 - e^{-4}(1 + 4 + 8) = 1 - 13e^{-4} \approx 0.762$, and the probability that the next quake is more than 2 weeks away is $e^{-4} \approx 0.018$ (whatever happened before now).

Example 9.2 (Location of an event)

Given $N(t) = 1$, the time T_1 of the event is uniform on $[0, t]$: for $s \le t$,

$$P(T_1 \le s \mid N(t) = 1) = \frac{P(N(s) = 1)P(N(t) - N(s) = 0)}{P(N(t) = 1)} = \frac{\lambda s e^{-\lambda s} e^{-\lambda(t-s)}}{\lambda t e^{-\lambda t}} = \frac{s}{t}.$$

More generally, given $N(t) = n$, the event times are distributed as the order statistics of n independent $U(0, t)$ variables.

Combining with Example 6.4: if each event of a rate-λ process is independently of type 1 with probability p, the type-1 and type-2 events form *independent* Poisson processes with rates λp and $\lambda(1 - p)$.

9.2 Markov chains

Motivation. Many systems evolve with no memory beyond the current state. Markov chains model this; their n-step probabilities are matrix powers, and in the long run they often settle into a stationary distribution regardless of the start.

Definition (Markov chain)

A sequence $X_0, X_1, \ldots$ with values in $\{0, 1, \ldots, M\}$ is a **Markov chain** if $P(X_{n+1} = j \mid X_n = i, X_{n-1} = i_{n-1}, \ldots, X_0 = i_0) = P_{ij}$ for all states and all n: the future depends on the past only through the present, with fixed **transition probabilities** $P_{ij} \ge 0$, $\sum_j P_{ij} = 1$.

The matrix $\mathbf{P} = (P_{ij})$ is the transition matrix, and

$$P(X_0 = i_0, \ldots, X_n = i_n) = P(X_0 = i_0) P_{i_0 i_1} \cdots P_{i_{n-1} i_n}.$$

Example 9.3

(a) Weather: if it rains today, it rains tomorrow with probability α; if not, with probability β. With state 0 = rain, $\mathbf{P} = \begin{pmatrix} \alpha & 1-\alpha \\ \beta & 1-\beta \end{pmatrix}$.
(b) Gambler's ruin: states $0, \ldots, N$, $P_{i,i+1} = p = 1 - P_{i,i-1}$ for $0 < i < N$, and $P_{00} = P_{NN} = 1$ (absorbing states).
(c) Ehrenfest urns: M molecules are distributed between two urns; at each step a molecule chosen at random changes urns. With X_n the number in urn 1, $P_{i,i-1} = i/M$ and $P_{i,i+1} = (M - i)/M$.
(d) If tomorrow's weather depends on the last two days, the pair (yesterday, today) is a Markov chain with four states: a process with longer memory can be made Markov by enlarging the state.

Proposition 9.3 (Chapman–Kolmogorov)

The n-step transition probabilities $P_{ij}^{(n)} = P(X_{m+n} = j \mid X_m = i)$ satisfy $P_{ij}^{(n+m)} = \sum_k P_{ik}^{(n)} P_{kj}^{(m)}$; hence $(P_{ij}^{(n)}) = \mathbf{P}^n$.

Proof. Condition on the state after n steps and use the Markov property. ∎

Example 9.4

In the weather chain with $\alpha = 0.7$, $\beta = 0.4$, the probability of rain four days after a rainy day is $(\mathbf{P}^4)_{00} = 0.5749$, and after a dry day it is $(\mathbf{P}^4)_{10} = 0.5668$: the initial state is already almost forgotten.

Theorem 9.4 (Limiting probabilities)

If for some n all entries of $\mathbf{P}^n$ are positive (the chain is *ergodic*), then $\pi_j = \lim_{n\to\infty} P_{ij}^{(n)}$ exists, does not depend on i, and is the unique solution of

$$\pi_j = \sum_i \pi_i P_{ij} \quad (j = 0, \dots, M), \qquad \sum_j \pi_j = 1.$$

The equations follow from $P_{ij}^{(n+1)} = \sum_k P_{ik}^{(n)} P_{kj}$ by letting $n \to \infty$ (existence of the limit is quoted). Thus π is a **stationary distribution**: if $X_0 \sim \pi$ then $X_n \sim \pi$ for all n. Moreover π_j is the long-run proportion of time spent in j, and $1/\pi_j$ is the mean number of steps between visits to j.

Example 9.5

(a) Weather: $\pi_0 = \alpha\pi_0 + \beta\pi_1$ and $\pi_0 + \pi_1 = 1$ give $\pi_0 = \beta/(1 + \beta - \alpha)$; for $\alpha = 0.7$, $\beta = 0.4$ this is $4/7 \approx 0.571$, matching Example 9.4.
(b) Ehrenfest: the binomial guess $\pi_j = \binom{M}{j} 2^{-M}$ satisfies $\pi_{j-1}\frac{M-j+1}{M} + \pi_{j+1}\frac{j+1}{M} = 2^{-M}\{\binom{M-1}{j-1} + \binom{M-1}{j}\} = \pi_j$, so in the long run each molecule is equally likely to be in either urn, independently. (This chain has period 2, so $P_{ij}^{(n)}$ itself does not converge, but π_j is still the long-run proportion of time in state j.)

9.3 Surprise, uncertainty and entropy

How surprised should we be to learn that an event of probability p occurred? Requiring the surprise $S(p)$ to be decreasing and continuous, with $S(1) = 0$ and $S(pq) = S(p) + S(q)$ (the surprise of two independent events adds), forces $S(p) = -C \log_2 p$; we take $C = 1$ and measure in **bits**. The functional equation is the one solved in Proposition 5.4.

The **entropy** of a discrete X with probabilities $p_1, \dots, p_n$ is the expected surprise

$$H(X) = -\sum_i p_i \log_2 p_i,$$

a measure of the uncertainty of X. It is 0 iff X is constant and at most $\log_2 n$, with equality iff X is uniform (by Jensen's inequality for the concave log: $\sum p_i \log(1/p_i) \le \log \sum p_i \cdot (1/p_i)$).

For a pair (X, Y), $H(X,Y) = -\sum_{i,j} p(x_i, y_j) \log_2 p(x_i, y_j)$, and the conditional entropy is $H_Y(X) = \sum_j H(X \mid Y = y_j) P(Y = y_j)$, the uncertainty remaining about X once Y is known. Then:

Proposition 9.5

(a) $H(X,Y) = H(Y) + H_Y(X)$.
(b) $H_Y(X) \le H(X)$, with equality iff X and Y are independent.

Proof. (a) Write $\log p(x_i,y_j) = \log p(y_j) + \log p(x_i \mid y_j)$ and take $-E$.

(b) Using $\log x \le x - 1$ (in natural logs),

$$H_Y(X) - H(X) = \sum_{i,j} p(x_i,y_j) \log_2 \frac{p(x_i)}{p(x_i \mid y_j)} \le \frac{1}{\ln 2} \sum_{i,j} p(y_j)\{p(x_i) - p(x_i \mid y_j)\} = 0,$$

with equality iff $p(x_i \mid y_j) = p(x_i)$ whenever $p(x_i,y_j) > 0$. ∎

So, on average, observing Y never increases our uncertainty about X, and $H(X) - H_Y(X)$ is the information Y carries about X.

9.4 Coding theory and entropy

We want to transmit the value of X (taking values $x_1,\dots,x_n$ with probabilities p_i) as a binary string, so that a sequence of messages can be decoded unambiguously. A code is a **prefix code** if no codeword is the beginning of another; then a concatenation is decoded by reading until a codeword is completed.

Lemma 9.6 (Kraft's inequality)

A prefix code with codeword lengths $n_1,\dots,n_N$ exists iff $\sum_i 2^{-n_i} \le 1$.

Proof. Let $L = \max n_i$. A codeword of length n_i is the prefix of 2^{L-n_i} strings of length L, and for a prefix code these sets are disjoint, so $\sum_i 2^{L-n_i} \le 2^L$. Conversely, if the inequality holds, assign codewords in order of increasing length, each time choosing a string not extending an earlier codeword; the count just given shows one is always available. ∎

Theorem 9.7 (Noiseless coding theorem)

Every prefix code for X has expected length $\sum_i p_i n_i \ge H(X)$, and there is a prefix code with expected length $< H(X) + 1$.

Proof. Let $q_i = 2^{-n_i}/\sum_k 2^{-n_k}$. By the same inequality as in Proposition 9.5(b), $-\sum p_i \log_2 p_i \le -\sum p_i \log_2 q_i = \sum p_i n_i + \log_2 \sum_k 2^{-n_k} \le \sum p_i n_i$ by Kraft. For the upper bound, take $n_i = \lceil \log_2(1/p_i) \rceil$: then $\sum 2^{-n_i} \le \sum p_i = 1$, so a prefix code exists, and $n_i < \log_2(1/p_i) + 1$. ∎

Coding blocks of k independent copies of X jointly brings the expected length per symbol within $1/k$ of $H(X)$: the entropy is the minimal number of bits per symbol needed.

Example 9.6

If X takes four values with probabilities $\frac12, \frac14, \frac18, \frac18$, then $H(X) = \frac12 + \frac24 + \frac38 + \frac38 = 1.75$ bits, and the prefix code $0, 10, 110, 111$ attains it exactly. A fixed-length code needs 2 bits.

Noisy channels. If each transmitted bit is flipped independently with probability p (binary symmetric channel), Shannon's theorem states that for every rate R below the **capacity** $C = 1 + p\log_2 p + (1-p)\log_2(1-p)$ bits per transmitted bit, there are codes whose error probability

tends to 0, while at rates above C this is impossible. Note that $C = 1 - H(\text{Bernoulli}(p))$: each received bit carries one bit minus the uncertainty introduced by the noise.

9.5 Selected exercises

Motivation. These are Ross's problems and theoretical exercises for Chapter 9 (book numbering): one computation with the Poisson process, a stationary distribution found by symmetry, and two consequences of the chain rule for entropy.

Exercise 9.1 (Two arrivals in the first hour)

Customers arrive at a bank as a Poisson process with rate λ. Given that two customers arrived during the first hour, find the probability that (a) both arrived during the first 20 minutes; (b) at least one did.

Solution.

Step 1. Measure time in hours and let $N(s)$ be the number of arrivals in $[0, s]$. By Proposition 9.1 and independent increments, $N(1/3)$ and $N(1) - N(1/3)$ are independent Poisson$(\lambda/3)$ and Poisson$(2\lambda/3)$, and $N(1) \sim$ Poisson(λ).

Step 2. For $k = 0, 1, 2$,

$$P(N(1/3) = k \mid N(1) = 2) = \frac{e^{-\lambda/3}\frac{(\lambda/3)^k}{k!}\, e^{-2\lambda/3}\frac{(2\lambda/3)^{2-k}}{(2-k)!}}{e^{-\lambda}\lambda^2/2} = \binom{2}{k}\left(\frac{1}{3}\right)^k\left(\frac{2}{3}\right)^{2-k},$$

a Bin$(2, \frac{1}{3})$ law, free of λ: each of the two arrivals behaves like an independent uniform time in the hour.

(a) $P(N(1/3) = 2 \mid N(1) = 2) = 1/9$.

(b) $P(N(1/3) \geq 1 \mid N(1) = 2) = 1 - (2/3)^2 = 5/9$. ■

Exercise 9.7 (Doubly stochastic chains)

A transition matrix on $\{0, 1, \ldots, M\}$ is doubly stochastic if $\sum_{i=0}^{M} P_{ij} = 1$ for every j. Show that if such a chain is ergodic, then $\pi_j = 1/(M+1)$ for all j.

Solution.

Step 1. Put $\pi_j = 1/(M+1)$. Then $\sum_j \pi_j = 1$ and, for every j,

$$\sum_{i=0}^{M} \pi_i P_{ij} = \frac{1}{M+1}\sum_{i=0}^{M} P_{ij} = \frac{1}{M+1} = \pi_j.$$

Step 2. By Theorem 9.4 the limiting probabilities of an ergodic chain are the unique solution of these equations, so they are the uniform probabilities. ■

Exercise 9.17 (Functions do not increase entropy)

Show that $H(f(X)) \leq H(X)$ for every discrete X and every function f.

Solution.

Step 1. Let $Y = f(X)$. The pair (X, Y) takes the value $(x, f(x))$ with probability $P(X = x)$ and every other value with probability 0, so $H(X, Y) = -\sum_x P(X = x) \log_2 P(X = x) = H(X)$.

Step 2. By Proposition 9.5(a), $H(X, Y) = H(Y) + H_Y(X)$. Each $H(X \mid Y = y)$ is the entropy of a distribution and so is nonnegative; hence $H_Y(X) \geq 0$.

Step 3. Combining, $H(Y) = H(X) - H_Y(X) \leq H(X)$, with equality iff $H_Y(X) = 0$, i.e. iff X is determined by $f(X)$ (f is one-to-one on the values X takes). ■

Exercise 9.18 (Capacity of the binary symmetric channel)

A bit X is sent and Y received, with $P(Y = 1 \mid X = 1) = P(Y = 0 \mid X = 0) = p$. The rate of transmission is $H(X) - H_Y(X)$ and the capacity is its maximum over $\alpha = P(X = 1)$. Show that the maximum is attained at $\alpha = \frac{1}{2}$ and equals $1 + p \log_2 p + (1 - p) \log_2 (1 - p)$.

Solution.

Step 1. Applying Proposition 9.5(a) in both orders, $H(X) + H_X(Y) = H(X, Y) = H(Y) + H_Y(X)$, so the rate equals $H(Y) - H_X(Y)$.

Step 2. Let $h(p) = -p \log_2 p - (1 - p) \log_2 (1 - p)$. Given $X = x$, Y equals x with probability p, so $H(Y \mid X = x) = h(p)$ for both values of x, and $H_X(Y) = h(p)$ whatever α is.

Step 3. Y takes two values, so $H(Y) \leq \log_2 2 = 1$, with equality iff $P(Y = 1) = \frac{1}{2}$. Since $P(Y = 1) = \alpha p + (1 - \alpha)(1 - p)$, the choice $\alpha = \frac{1}{2}$ gives $P(Y = 1) = \frac{1}{2}$.

Step 4. Hence the rate $H(Y) - h(p)$ is at most $1 - h(p)$, with equality at $\alpha = \frac{1}{2}$, and the capacity is $1 - h(p) = 1 + p \log_2 p + (1 - p) \log_2 (1 - p)$, the value C quoted at the end of Section 9.4. ■

Lecture 10. Simulation

10.1 Random numbers and simulation

When a probability or expectation is too complicated to compute, we can estimate it by **simulation**: generate many independent realizations of the experiment and use the strong law of large numbers. Everything is built from **random numbers**, independent $U(0,1)$ variables (in practice produced by deterministic pseudorandom generators). For instance, $\lfloor nU \rfloor + 1$ is uniform on $\{1, \dots, n\}$.

Example 10.1 (A random permutation)

To generate a uniformly random permutation of $1, \dots, n$: start from any ordering $P_1, \dots, P_n$; for $k = n, n-1, \dots, 2$, choose I uniform on $\{1, \dots, k\}$ and swap P_I and P_k. Position n receives each element with probability $1/n$, then position $n-1$ each remaining element with probability $1/(n-1)$, and so on, so every permutation has probability $1/n!$. The first r positions filled give a random r-subset, which is how subjects are randomly assigned to treatments.

10.2 Simulating continuous random variables

Motivation. Computers produce uniform random numbers; every other distribution must be built from them. The inverse transform and rejection methods are the two general techniques.

10.2.1 The inverse transformation method

Proposition 10.1

If F is a continuous cdf and $U \sim U(0,1)$, then $X = F^{-1}(U)$ has cdf F.

Proof. $P(F^{-1}(U) \le x) = P(U \le F(x)) = F(x)$, since F is increasing. ∎

Example 10.2

(a) Exponential: $F(x) = 1 - e^{-\lambda x}$ gives $X = -\frac{1}{\lambda}\log(1-U)$, or equivalently $-\frac{1}{\lambda}\log U$, since $1-U$ is also uniform.
(b) Gamma(n, λ) for integer n: add n independent exponentials, $X = -\frac{1}{\lambda}\log(U_1 U_2 \cdots U_n)$, which needs only one logarithm.

10.2.2 The rejection method

Suppose we can simulate from a density g, and $f(y)/g(y) \le c$ for all y.

Theorem 10.2 (Rejection method)

Repeat: generate $Y \sim g$ and an independent $U \sim U(0,1)$, until $U \le f(Y)/\{cg(Y)\}$; then set $X = Y$. The resulting X has density f, and the number of iterations is geometric with mean c.

Proof. In one iteration, $P(\text{accept}) = \int P(U \le \frac{f(y)}{cg(y)}) g(y)\,dy = \int \frac{f(y)}{c}\,dy = \frac{1}{c}$, and

$$P(Y \le x \mid \text{accept}) = c \int_{-\infty}^{x} \frac{f(y)}{cg(y)} g(y)\,dy = F(x).$$

Iterations are independent, so the number needed is geometric with success probability $1/c$, and the accepted value has the conditional law just computed. ■

Example 10.3 (Normal from exponentials)

$|Z|$ has density $f(x) = \frac{2}{\sqrt{2\pi}} e^{-x^2/2}$, $x > 0$. With $g(x) = e^{-x}$,

$$\frac{f(x)}{g(x)} = \sqrt{\frac{2}{\pi}}\, e^{x - x^2/2} \le \sqrt{\frac{2e}{\pi}} = c \approx 1.32,$$

the maximum being at $x = 1$, and $\frac{f(x)}{cg(x)} = \exp\{-(x-1)^2/2\}$. So: generate $Y = -\log U_1$, accept if $U_2 \le e^{-(Y-1)^2/2}$ (equivalently if an independent exponential exceeds $(Y-1)^2/2$), and finally attach a random sign. About 1.32 iterations are needed per normal.

Example 10.4 (The polar method)

By Example 6.14, if $R^2 \sim \text{Exp}(1/2)$ and $\Theta \sim U(0, 2\pi)$ are independent, then $R\cos\Theta$, $R\sin\Theta$ are independent standard normals. To avoid computing sines and cosines, generate $V_i = 2U_i - 1$ $(i = 1, 2)$ until $S = V_1^2 + V_2^2 \le 1$; then (V_1, V_2) is uniform on the unit disc, S is uniform and independent of the angle, and $\cos\Theta = V_1/\sqrt{S}$, $\sin\Theta = V_2/\sqrt{S}$. With $R^2 = -2\log S$,

$$X = V_1\sqrt{\frac{-2\log S}{S}}, \qquad Y = V_2\sqrt{\frac{-2\log S}{S}}$$

are independent standard normals. The acceptance probability is $\pi/4$.

A χ^2_n variable can be simulated as $\sum Z_i^2$, or for even n as $-2\log(U_1 \cdots U_{n/2})$ (a Gamma$(n/2, 1/2)$), adding one Z^2 for odd n.

10.3 Simulating from discrete distributions

For $P(X = x_j) = p_j$, set $X = x_j$ if $\sum_{i<j} p_i \le U < \sum_{i \le j} p_i$; this happens with probability p_j. It is the discrete inverse transform.

Example 10.5

(a) Geometric: $X = j$ iff $1 - q^{j-1} \le U < 1 - q^j$ $(q = 1 - p)$, i.e. $X = 1 + \lfloor \log U / \log q \rfloor$ (using $1 - U$ in place of U).
(b) Binomial: $X = \sum_{i=1}^{n} \mathbb{1}\{U_i < p\}$.
(c) Poisson(λ): with $N(1)$ the number of events of a rate-λ process in $[0, 1]$, and interarrival times $-\log U_i/\lambda$,

$$N = \max\Big\{n : \prod_{i=1}^{n} U_i \ge e^{-\lambda}\Big\}$$

is Poisson(λ): multiply random numbers until the product drops below $e^{-\lambda}$, and subtract 1 from the count.

10.4 Variance reduction

To estimate $\theta = E[X]$ we average n independent copies, with error of order $\sqrt{\mathrm{Var}(X)/n}$. Reducing the variance of what is averaged is as good as increasing n.

10.4.1 Antithetic variables

If X_1, X_2 have the same distribution, $\mathrm{Var}(\frac{X_1+X_2}{2}) = \frac{1}{4}\{2\,\mathrm{Var}(X_1) + 2\,\mathrm{Cov}(X_1, X_2)\}$, which is smaller than for independent copies when $\mathrm{Cov}(X_1, X_2) < 0$. If $X_1 = h(U_1, \dots, U_m)$ with h monotone in each argument, then $X_2 = h(1 - U_1, \dots, 1 - U_m)$ has the same distribution and (as can be shown) $\mathrm{Cov}(X_1, X_2) \le 0$. So each set of random numbers is used twice, once "reversed".

10.4.2 Variance reduction by conditioning

By the conditional variance formula (Proposition 7.6), $\mathrm{Var}(E[X \mid Y]) \le \mathrm{Var}(X)$, and $E[X \mid Y]$ has the same mean θ. So, if $E[X \mid Y]$ can be computed, simulate Y and average $E[X \mid Y]$ instead of X.

Example 10.6 (Estimating π)

With V_1, V_2 independent $U(-1, 1)$ and $I = \mathbb{1}\{V_1^2 + V_2^2 \le 1\}$, $E[I] = \pi/4$ and $\mathrm{Var}(I) = \frac{\pi}{4}(1 - \frac{\pi}{4}) \approx 0.1685$. Conditioning on V_1,

$$E[I \mid V_1] = P(V_2^2 \le 1 - V_1^2 \mid V_1) = \sqrt{1 - V_1^2},$$

so we may average $\sqrt{1 - U^2}$ over random numbers U (the sign of V_1 is irrelevant): $\mathrm{Var}(\sqrt{1 - U^2}) = \frac{2}{3} - \frac{\pi^2}{16} \approx 0.0498$, a 3.4-fold reduction. Since $\sqrt{1 - u^2}$ is monotone, antithetic pairs $\frac{1}{2}\{\sqrt{1 - U^2} + \sqrt{1 - (1 - U)^2}\}$ do better still: their variance is about 0.0069 per pair, against 0.0249 for two independent values.

10.4.3 Control variates

Suppose Y is simulated alongside X and $\mu_Y = E[Y]$ is known. Then $X + c(Y - \mu_Y)$ also has mean θ, and its variance $\mathrm{Var}(X) + c^2\,\mathrm{Var}(Y) + 2c\,\mathrm{Cov}(X, Y)$ is minimized by

$$c^* = -\frac{\mathrm{Cov}(X, Y)}{\mathrm{Var}(Y)}, \qquad \mathrm{Var}(X + c^*(Y - \mu_Y)) = \mathrm{Var}(X)(1 - \rho(X, Y)^2).$$

The more strongly Y is correlated with X, the larger the gain; in practice c^* is estimated from the simulated pairs. In Example 10.6, $Y = U^2$ with $E[Y] = 1/3$ is a natural control variate for $\sqrt{1 - U^2}$.

10.5 Selected exercises

Motivation. Exercises 10.1–10.13 are Ross's Chapter 10 problems (book numbering); they prove that the simulation recipes of this lecture are correct or optimal. Exercise 10.A is added to justify the claim, made above without proof, that antithetic pairs are negatively correlated.

Exercise 10.1 (Random permutations by insertion)

Set $P(1) = 1$. For $k = 2, \dots, n$, generate a random number U, let $J = \lfloor kU \rfloor + 1$, and set $P(k) = P(J)$ and then $P(J) = k$ (so for $J = k$ simply $P(k) = k$). Show that after step k, $(P(1), \dots, P(k))$ is a uniformly random permutation of $1, \dots, k$.

Solution.

Step 1. In step k the algorithm puts k into position J and moves the element previously at position J to the new position k; J is uniform on $\{1, \dots, k\}$ and independent of the earlier steps.

Step 2. Let Φ map a pair (σ, j), with σ a permutation of $1, \dots, k-1$ and $j \in \{1, \dots, k\}$, to the result τ of this operation. Φ is a bijection onto the permutations of $1, \dots, k$: given τ, j is the position of k in τ, and σ is recovered by putting $\tau(k)$ back into position j (if $j < k$) and deleting position k.

Step 3. (Induction.) After step 1 the claim is trivial. If after step $k-1$ each σ has probability $1/(k-1)!$, then by independence each pair (σ, j) has probability $\frac{1}{(k-1)!} \cdot \frac{1}{k}$, and by Step 2 each permutation τ of $1, \dots, k$ arises from exactly one pair. So each τ has probability $1/k!$. ■

Exercise 10.7 (The maximum of random numbers)

Let $F(x) = x^n$ for $0 < x < 1$.
(a) Simulate from F using a single random number.
(b) Show that $\max(U_1, \dots, U_n)$ has distribution F for independent random numbers $U_1, \dots, U_n$, and so give a second method.

Solution.

(a) F is continuous and strictly increasing on $(0, 1)$ with $F^{-1}(u) = u^{1/n}$, so $X = U^{1/n}$ has distribution F by Proposition 10.1: $P(U^{1/n} \le x) = P(U \le x^n) = x^n$.

(b) $\max_i U_i \le x$ iff every $U_i \le x$, so by independence $P(\max_i U_i \le x) = \prod_i P(U_i \le x) = x^n$. Thus $\max(U_1, \dots, U_n)$ is a second (more expensive) way to generate from F. ■

Exercise 10.10 (Choosing the exponential envelope)

In Example 10.3, use the envelope $g_\lambda(x) = \lambda e^{-\lambda x}$, $x > 0$, instead of e^{-x}. Show that the mean number of iterations of the rejection method is minimized when $\lambda = 1$.

Solution.

Step 1. With $f(x) = \sqrt{2/\pi}\, e^{-x^2/2}$,

$$\frac{f(x)}{g_\lambda(x)} = \frac{1}{\lambda}\sqrt{\frac{2}{\pi}}\, e^{\lambda x - x^2/2} = \frac{1}{\lambda}\sqrt{\frac{2}{\pi}}\, e^{\lambda^2/2} e^{-(x-\lambda)^2/2},$$

whose maximum over $x > 0$ is attained at $x = \lambda$. So the smallest admissible constant is $c(\lambda) = \sqrt{2/\pi}\, e^{\lambda^2/2}/\lambda$.

Step 2. By Theorem 10.2 the mean number of iterations is $c(\lambda)$. Now $\frac{d}{d\lambda} \log c(\lambda) = \lambda - \frac{1}{\lambda}$ is negative on $(0, 1)$ and positive on $(1, \infty)$, so c is minimized at $\lambda = 1$, with $c(1) = \sqrt{2e/\pi} \approx 1.32$. ■

Exercise 10.13 (Polar coordinates of a uniform point in the disc)

Let (X, Y) be uniform on the unit disc, with density $1/\pi$ there, and let (R, Θ) be its polar coordinates, $\Theta \in (0, 2\pi)$.

(a) Show that R and Θ are independent, $R^2 \sim U(0,1)$ and $\Theta \sim U(0, 2\pi)$.
(b) Show that if V_1, V_2 are independent $U(-1,1)$, then given $V_1^2 + V_2^2 \le 1$ the point (V_1, V_2) is uniform on the unit disc. (Together, (a) and (b) justify the polar method, Example 10.4.)

Solution.

(a)

Step 1. The map $(r,\theta) \mapsto (r\cos\theta, r\sin\theta)$ is one-to-one from $(0,1)\times(0,2\pi)$ onto the disc minus a segment of probability 0, with Jacobian r. By Theorem 6.3 applied to its inverse, (R,Θ) has density

$$f_{R,\Theta}(r,\theta) = \frac{1}{\pi}\cdot r = 2r\cdot\frac{1}{2\pi}, \qquad 0<r<1,\ 0<\theta<2\pi.$$

Step 2. The density factors, so R and Θ are independent (Proposition 6.1), with densities $2r$ on $(0,1)$ and $1/(2\pi)$ on $(0,2\pi)$. Finally $P(R^2 \le s) = P(R \le \sqrt{s}) = \int_0^{\sqrt{s}} 2r\,dr = s$ for $0<s<1$.

(b) (V_1, V_2) is uniform on the square $[-1,1]^2$ of area 4, and the disc D has area π. For $A \subseteq D$,

$$P((V_1,V_2)\in A \mid (V_1,V_2)\in D) = \frac{\text{area}(A)/4}{\pi/4} = \frac{\text{area}(A)}{\pi},$$

which is the uniform distribution on D. ■

Exercise 10.A (Antithetic variables are negatively correlated)

Let h be nondecreasing on $(0,1)$ with $E[h(U)^2] < \infty$ for $U \sim U(0,1)$. Show that $\mathrm{Cov}(h(U), h(1-U)) \le 0$.

Solution. [Supplement]

Step 1. Put $k(u) = h(1-u)$, which is nonincreasing. For all $u, v \in (0,1)$ the numbers $h(u) - h(v)$ and $k(u) - k(v)$ have opposite signs (or one is 0), so

$$\big(h(u)-h(v)\big)\big(k(u)-k(v)\big) \le 0.$$

Step 2. Let $V \sim U(0,1)$ be independent of U and substitute $u = U$, $v = V$. Taking expectations, expanding, and using that U and V have the same law and are independent,

$$0 \ge E[h(U)k(U)] - E[h(U)]E[k(V)] - E[h(V)]E[k(U)] + E[h(V)k(V)] = 2\,\mathrm{Cov}\big(h(U),k(U)\big).$$

All expectations are finite by the Cauchy–Schwarz inequality, since $k(U) = h(1-U)$ has the same law as $h(U)$.

Step 3. Hence $\mathrm{Cov}(h(U), h(1-U)) \le 0$, and the antithetic estimator $\frac{1}{2}\{h(U) + h(1-U)\}$ has variance at most $\frac{1}{2}\mathrm{Var}(h(U))$, the variance of the average of two independent copies. ■